

Super interesting title on weakly interacting fermionic systems

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1 Second Quantization

Let's take a step back and build the correct structures in which to set the treatment of this physical problem.

1.1 Principles of quantum mechanics

Preliminaries

We must start our discussion of the tools needed to approach the physical problem at hand by defining the mathematical objects that appear in the description of a quantum system. The probabilistic nature of a quantum system is realized in the linear structure of a complex vector space that is equipped with an inner product, which produces the probability amplitude for a system prepared in

some state to be measured in another. In this setting, linear combinations of state vectors are just probabilistic superpositions.

We will give the necessary definitions regarding a particular class of vector spaces, even though many concepts come from much more general settings.

Definition 1.1 (Hilbert space). A (complex) *Hilbert space* $\mathcal{H}$ is a vector space equipped with a sesquilinear map $\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ such that $\|\phi\| := \langle \phi, \phi \rangle^{1/2}$ defines a norm which makes $\mathcal{H}$ a complete metric space.

A note on the choice of notation. Usually a vector is just indicated by a letter, but there are situations in quantum mechanics where it is clearer to indicate a family of vectors by an index. To prevent ambiguity in those cases it can be convenient to employ Dirac's "bra-ket" notation, where a vector would be indicated by $|\psi\rangle \in \mathcal{H}$, a linear functional as $\langle\psi| \in \mathcal{H}^*$, and the inner product as $\langle\psi, A\phi\rangle = \langle\psi|A|\phi\rangle$. It will be clear by the context if this notation is used.

Definition 1.2. A Hilbert space is said to be *separable* if it admits a countable dense subset.

Even when writing just "Hilbert space" we will always assume that the space is separable. Observable quantities must then be represented by objects that are capable of encoding statistical information about the system. In particular, the result of an experiment is an average of all the possible outcomes allowed by the state the system is in, and this average must be a real number. If $\{\lambda_i : i \in I\} \subseteq \mathbb{R}$ is the set of the possible values that can be obtained by measuring a physical quantity A , and the system is found in a superposition of the final states $\{\psi_i : i \in I\}$ on which the system could collapse after the measurement, then we would want to write the expectation value as the result of applying a linear map on this state

$$\langle A \rangle_{\psi_0} = \sum_{i \in I} \lambda_i \langle \psi_i, \psi_i \rangle = \langle \psi, A\psi \rangle$$

and this suggests that the natural definition for an observable is a *symmetric operator*, which we can now define, with measurable outcomes being its eigenvalues.

Definition 1.3 (Operator). A *linear operator* (or simply an *operator*) A on $\mathcal{H}$ is a linear map $A : D(A) \rightarrow \mathcal{H}$ defined on a subspace $D(A) \subseteq \mathcal{H}$. If $D(A)$ is dense in $\mathcal{H}$, A is said to be *densely defined*.

For the sake of brevity, we will say "operator" to mean a densely defined linear map on a Hilbert space, unless stated otherwise. Moreover, note that the domain on which an operator acts is part of the definition, and will therefore need to be specified case by case.

Definition 1.4 (Extension of an operator). If A and B are two operators such that $D(A) \subseteq D(B)$ and $A\phi = B\phi$ for all $\phi \in D(A)$, then we say that B *extends* (or is an extension of) A , and we write $A \subseteq B$.

Definition 1.5 (Observable). An *observable* is a symmetric operator on $\mathcal{H}$, i.e. such that $\langle\psi, A\psi\rangle = \langle A\psi, \psi\rangle$ for all $\psi \in \mathcal{H}$.

Remark 1. The symmetry condition $\langle\psi, A\phi\rangle = \langle A\psi, \phi\rangle$ for all $\phi, \psi \in \mathcal{H}$ is equivalent to the condition $\langle\psi, A\psi\rangle \in \mathbb{R}$ for all $\psi \in \mathcal{H}$, and because we defined expectation values as exactly this type of expression, this is equivalent to asking that measurable quantities must be real numbers.

Definition 1.6 (Bounded operator). An operator A is said to be *bounded* if $D(A) = \mathcal{H}$ and is continuous (with respect to the norm topology), which by linearity is equivalent to the condition

$$\|A\| := \sup_{\psi, \|\psi\|=1} \|A\psi\| < \infty.$$

The quantity $\|A\|$ is called the *operator norm* of A . An operator is said to be *unbounded* if it is not bounded.

Proposition 1.1. *Given $A : D(A) \rightarrow \mathcal{H}$ a densely defined bounded operator on $\mathcal{H}$, there exists a unique extension to a bounded operator on the entire space $\tilde{A} : \mathcal{H} \rightarrow \mathcal{H}$.*

Proof. Let $\{x_n\}_{n \in \mathbb{N}} \subseteq D(A)$ be a Cauchy sequence in $D(A)$, then there exists a unique $x \in \mathcal{H}$ such that $x_n \rightarrow x$. Because A is bounded it holds that $\|Ax_n - Ax_m\| \leq \|A\| \|x_n - x_m\|$ and therefore the sequence $\{Ax_n\}_{n \in \mathbb{N}}$ is also a Cauchy sequence. As such, there exists a unique $y \in \mathcal{H}$ such that $Ax_n \rightarrow y$. Define the operator $\tilde{A}$ as

$$\begin{aligned}\tilde{A} : \mathcal{H} &\longrightarrow \mathcal{H} \\ x &\longmapsto \tilde{A}x := \lim_n Ax_n\end{aligned}$$

where $\{x_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence that converges to x . $\tilde{A}$ extends A and is defined on the entire space. Moreover, it is bounded as $\|\tilde{A}\| < \infty$ and

$$\|\tilde{A}x\| = \|\lim_n Ax_n\| = \lim_n \|Ax_n\| \leq \lim_n \|A\| \|x_n\| = \|A\| \|x\|.$$

□

From this simple but important result it is clear that, when dealing with bounded operators, it will suffice to define said operator on a dense subspace. In particular, since we are always assuming the Hilbert space to be separable, it will suffice to define the image of only the basis elements, i.e. the elements of the appropriate countable dense subset.

Definition 1.7 (Adjoint). If A is an operator on $\mathcal{H}$ we define the *adjoint* A^* of A as the operator $A^* : D(A^*) \rightarrow \mathcal{H}$ defined on

$$D(A^*) = \{\phi \in \mathcal{H} : \sup_{\psi \in D(A), \|\psi\|=1} |\langle \phi, A\psi \rangle| < \infty\}$$

and with $A^*\phi$ defined for all $\phi \in \mathcal{H}$ as

$$\langle A^*\phi, \psi \rangle = \langle \phi, A\psi \rangle \quad \forall \psi \in D(A).$$

Proposition 1.2. *The adjoint of a bounded operator is a bounded operator.*

Proof. ????

□

Proposition 1.3. *A is a symmetric operator if and only if A^* is an extension of A , i.e. $A \subseteq A^*$.*

Proof. ????

□

Definition 1.8 (Compact, trace class and Hilbert-Schmidt operators). An operator A is said to be *compact* if $D(A) = \mathcal{H}$ and there are orthonormal bases $\{u_n\}_{n \in \mathbb{N}}$ and $\{v_n\}_{n \in \mathbb{N}}$ and a sequence $\{\lambda_n\}_{n \in \mathbb{N}} \subset \mathbb{C}$ with $\lim_n \lambda_n = 0$ such that

$$A\phi = \sum_{n=0}^{\infty} \lambda_n \langle v_n, \phi \rangle u_n.$$

A compact operator A is said to be *trace class* if $\sum_{n=1}^{\infty} |\lambda_n| < \infty$ and is said to be *Hilbert-Schmidt* if $\sum_{n=1}^{\infty} |\lambda_n|^2 < \infty$. For a trace class operator A we define its *trace* as

$$\text{Tr}(A) := \sum_{n=1}^{\infty} \lambda_n \langle v_n, u_n \rangle.$$

Proposition 1.4. *The trace of a trace class operator A is finite, and given an orthonormal basis $\{\phi_n\}_{n \in \mathbb{N}}$ for $\mathcal{H}$ the trace is*

$$\text{Tr}(A) = \sum_{n=1}^{\infty} \langle \phi_n, A\phi_n \rangle.$$

Proof. ????

□

We now present some important theorems for which we will not give the proof, as they go beyond the scope of our work. Recall that the dual of a vector space is the set of all the *continuous* linear functionals on that space, where the topology that defines continuity is once again the norm topology.

MAYBE THESE THEOREMS COULD BE PUT IN AN APPENDIX?

Theorem 1.5 (Riesz's representation theorem). *Given a linear map $F : \mathcal{H} \rightarrow \mathbb{C}$, $F \in \mathcal{H}^*$ if, and only if, there exists a unique $x_F \in \mathcal{H}$ such that $F(x) = \langle x_F, x \rangle$ for all $x \in \mathcal{H}$. Furthermore, $\|F\|_{\mathcal{H}^*} = \|x_F\|_{\mathcal{H}}$.*

Proof. AAAAAAAHHHHHHHHH.

□

Theorem 1.6 (Spectral theorem for compact operators). *Given a compact symmetric operator A , there exists an orthonormal basis $\{\phi_n\}_{n \in \mathbb{N}}$ for $\mathcal{H}$ and a sequence $\{\lambda_n\}_{n \in \mathbb{N}}$ such that $\lim_n \lambda_n = 0$ and*

$$A\psi = \sum_{n=1}^{\infty} \lambda_n \langle \phi_n, \psi \rangle \phi_n \quad \forall \psi \in \mathcal{H}.$$

The set $\{\lambda_n\}_n$ is the point spectrum of A and $\{\phi_n\}_n$ are its eigenvectors, which also form an orthonormal basis for $\mathcal{H}$.

The most important Hilbert space for us is the space of square integrable functions on $\mathbb{R}^n$, i.e.

$$L^2(\mathbb{R}^n) = \left\{ f : \mathbb{R}^n \rightarrow \mathbb{C} \mid \left(\int_{\mathbb{R}^n} dx |f(x)|^2 \right)^{1/2} < \infty \right\},$$

whose elements are the wavefunctions of a physical system.

Going back to the definition of expectation value of an observable, we can now make an important generalization. Suppose the system has probability p_i of being prepared in some initial state ψ_i , of course given the probability normalization condition $\sum_i p_i = 1$, then the expectation of an observable A would be the weighted average of the expectations on each initial state.

Definition 1.9 (Pure and mixed states). A state represented by a vector in $\mathcal{H}$ is called a *pure state*, while the statistical mixture of pure states $\{(p_i, \psi_i) : i \in I\}$, with $\sum_i p_i = 1$, is called a *mixed state*. If A is an observable, the expectation of A on the mixed state $\{(p_i, \psi_i)\}$ is defined as the average of the expectations over each single state

$$\langle A \rangle = \sum_i p_i \langle \psi_i, A\psi_i \rangle$$

Assuming the observable A is a **compact** symmetric operator, by the spectral theorem we can write it as a series of projectors onto its eigenspaces. Let $\{\phi_\lambda : \lambda \in \sigma(A)\}$ be the eigenvectors of A , then

$$A = \sum_{\lambda \in \sigma(A)} \lambda \langle \phi_\lambda, \cdot \rangle \phi_\lambda.$$

Using this, we write the expectation of A on the mixed state $\{(p_i, \psi_i)\}$ as

$$\langle A \rangle = \sum_{i \in I} p_i \langle \psi_i, A \psi_i \rangle = \sum_{i \in I, \lambda \in \sigma(A)} p_i \langle \psi_i, \lambda \langle \phi_\lambda, \psi_i \rangle \phi_\lambda \rangle = \sum_{\lambda \in \sigma(A)} \lambda \langle \phi_\lambda, \rho \phi_\lambda \rangle = \text{Tr}(\rho A)$$

where in the last step we defined the *density matrix* which uniquely identifies the mixed state

$$\rho := \sum_i p_i \langle \psi_i, \cdot \rangle \psi_i.$$

This brings us to a general definition of a quantum state. Recall that $\mathfrak{B}(\mathcal{H})$ is the C^* -algebra of bounded operators on $\mathcal{H}$.

Definition 1.10. A general quantum state is an operator ρ on $\mathcal{H}$ such that

- $\text{Tr}(\rho) = 1$ (probability normalization condition)
- $\rho \geq 0$ (positivity of probabilities)
- $\rho^* = \rho$ (symmetry)

and defines the expectation functional

$$\begin{aligned} \langle \cdot \rangle : \mathfrak{B}(\mathcal{H}) &\longrightarrow \mathbb{C} \\ A &\longmapsto \langle A \rangle := \text{Tr}(\rho A). \end{aligned}$$

Equipped with this general definition we are now ready to define the fundamental thermodynamic quantity that will be the center of the rest of our work, the free energy.

Definition 1.11 (Von Neumann entropy). In the same notation used so far (and with $I = \mathbb{N}$), the *Von Neumann entropy* of a state $\langle \cdot \rangle$ is defined as

$$S(\langle \cdot \rangle) := - \sum_{n=1}^{\infty} p_n \log(p_n),$$

where we are using the convention that $t \log(t) = 0$ if $t = 0$.

We are particularly interested in *equilibrium states*. Zero temperature states are usually pure, while non-zero temperature are mixed.

Definition 1.12 (Stability and ground state). Given a physical system governed by a Hamiltonian H on $\mathcal{H}$, if

$$\inf_{\phi \in D(H), \|\phi\|=1} \langle \phi, H \phi \rangle > -\infty$$

the system is said to be *stable*. We define the *ground state energy* as

$$E = \inf_{\phi \in D(H), \|\phi\|=1} \langle \phi, H \phi \rangle.$$

A *ground state, if it exists*, is a unit vector $\psi_0 \in D(H)$ such that

$$\langle \psi_0, H \psi_0 \rangle = E,$$

i.e. a minimizer for the energy expectation functional $\langle \cdot, H \cdot \rangle$ we just defined.

Remark 2. If the ground state ψ_0 exists and the ground state energy is E then it holds that $H\psi_0 = E\psi_0$, i.e. ψ_0 is an eigenvector of the energy operator with eigenvalue equal to the ground state energy.

Definition 1.13 (Free energy). The *free energy* of a stable system at temperature $T \geq 0$ is defined as

$$F(T) := \inf_{\langle \cdot \rangle, \langle H \rangle < \infty} (\langle H \rangle - T S(\langle \cdot \rangle)),$$

which is possibly $-\infty$. Note that we are also assuming that the states ψ_i in the definition of a mixed state are in $D(H)$. If, for $T > 0$, there exists a minimizer for the free energy variation, then that state is called a *Gibbs state at temperature T* .

Remark 3. A system composed of a single free particle is stable but it does not admit a ground state, and $F(T) = -\infty$ for all $T > 0$.

Proposition 1.7. If $\langle \cdot \rangle$ is a Gibbs state at $T > 0$, then for all bounded operators

$$\langle A \rangle = \frac{\text{Tr}(A \exp(-\beta H))}{\text{Tr}(\exp(-\beta H))}$$

where $\beta := 1/T$. In particular, $\exp(-\beta H)$ is a trace class operator.

Proof. ??? □

Tensor product of Hilbert spaces

Add discussion about direct sum of hilbert spaces and prove that direct sums and tensor products of hilbert spaces are still hilbert spaces?

So far we talked about a generic (separable) Hilbert space, but for the most part we will deal with Hilbert spaces constructed in a specific way: tensor products of Hilbert spaces, which are an especially useful construction as they are the natural spaces in which to set physical systems composed of multiple particles. In particular, we are interested in two special subspaces of a tensor product that correspond, as we will see, to the space of actual physical states for bosonic and fermionic systems, and we will naturally be interested in restricting operators to those subspaces via projections.

Definition 1.14 (Tensor product Hilbert space). Given $\mathcal{H}$ and $\mathcal{K}$ two Hilbert spaces, their *tensor product* is a Hilbert space, denoted $\mathcal{H} \otimes \mathcal{K}$, together with a bilinear map

$$\mathcal{H} \times \mathcal{K} \ni (u, v) \mapsto u \otimes v \in \mathcal{H} \otimes \mathcal{K}$$

such that the inner product satisfies

$$\langle u_1 \otimes v_1, u_2 \otimes v_2 \rangle_{\mathcal{H} \otimes \mathcal{K}} = \langle u_1, u_2 \rangle_{\mathcal{H}} \langle v_1, v_2 \rangle_{\mathcal{K}}$$

for all $u_1, u_2 \in \mathcal{H}$ and $v_1, v_2 \in \mathcal{K}$, and such that the space $\text{span}\{u \otimes v : u \in \mathcal{H}, v \in \mathcal{K}\}$ is dense in $\mathcal{H} \otimes \mathcal{K}$.

Remark 4. One can show that the tensor product between Hilbert spaces is unique up to isometric isomorphism (i.e. up to an isomorphism that preserves the inner product). Moreover, given orthonormal bases $\{u_i\}_{i \in I}$ for $\mathcal{H}$ and $\{v_j\}_{j \in J}$ for $\mathcal{K}$, the set $\{u_n \otimes v_m\}_{n,m}$ is a basis for the space $\mathcal{H} \otimes \mathcal{K}$, which can be naturally identified with the space $\ell^2(I \times J)$ where the inner product acts as

$$(u \otimes v)_{ij} = \langle u_i, u \rangle_{\mathcal{H}} \langle v_j, v \rangle_{\mathcal{K}}.$$

Remark 5. The space $L^2(\mathbb{R}^n) \otimes L^2(\mathbb{R}^m)$ can be naturally identified with the space $L^2(\mathbb{R}^{n+m})$, and given $f \in L^2(\mathbb{R}^n)$ and $g \in L^2(\mathbb{R}^m)$ one has

$$f \otimes g(x, y) = f(x)g(y).$$

Given two operators A on $\mathcal{H}$ and B on $\mathcal{K}$ we can define an operator $A \otimes B$ on $\mathcal{H} \otimes \mathcal{K}$ in a natural way. Its domain will be

$$D(A \otimes B) = \{\phi \otimes \psi : \phi \in D(A), \psi \in D(B)\},$$

which is a dense subspace of the tensor product by construction, and on this subspace it will act as

$$A \otimes B(\phi \otimes \psi) = (A\phi) \otimes (B\psi).$$

We can extend the definition of a tensor product to products of N spaces in a very natural way by substituting pure tensors with N -fold pure tensors and inner products with N -fold products of the respective inner products. This way, the definition given is just the special case $N = 2$. Of special interest to us is the N -fold tensor product of $\mathcal{H}$ with itself $\bigotimes^N \mathcal{H}$, on which we can define a natural action for the group of permutations S_N . Indeed, we can define the unitary representation U_σ of $\sigma \in S_N$ on this space as

$$U_\sigma(u_1 \otimes \cdots \otimes u_N) := u_{\sigma^{-1}(1)} \otimes \cdots \otimes u_{\sigma^{-1}(N)}.$$

With this action, we define two important subspaces. Define the operators

$$\begin{aligned} P_+ &:= \frac{1}{N!} \sum_{\sigma \in S_N} U_\sigma \\ P_- &:= \frac{1}{N!} \sum_{\sigma \in S_N} \text{sgn}(\sigma) U_\sigma. \end{aligned} \tag{1}$$

These operators are unitary by construction and **one can show** that they are also projections.

Definition 1.15 (Symmetric and anti-symmetric tensor products). The *symmetric tensor product* is the subspace

$$\bigotimes_{sym}^N \mathcal{H} := P_+ \left(\bigotimes^N \mathcal{H} \right)$$

and the *anti-symmetric tensor product* is the subspace

$$\bigwedge^N \mathcal{H} := P_- \left(\bigotimes^N \mathcal{H} \right).$$

We also define the anti-symmetric tensor product of vectors as

$$u_1 \wedge \cdots \wedge u_N = \sqrt{N!} P_- (u_1 \otimes \cdots \otimes u_N).$$

Remark 6. In the special case where $\mathcal{H} = L^2(\mathbb{R}^n)$, anti-symmetric tensor products between wave-functions can be constructed as Slater determinants, i.e.

$$u_1 \wedge \cdots \wedge u_N(x_1, \dots, x_N) = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} u_1(x_1) & \cdots & u_N(x_1) \\ \vdots & & \vdots \\ u_1(x_N) & \cdots & u_N(x_N) \end{pmatrix}$$

Remark 7. If $\dim(\mathcal{H}) < N$ then $\bigwedge^N \mathcal{H} = \{0\}$, i.e. the anti-symmetrization of N vectors in a space of dimension less than N is always the null vector. This is especially important because physical states of fermionic systems live in the anti-symmetric tensor product, and if the Hilbert space of a single particle is finite-dimensional the allowed number of particles will be finite.

1.2 Canonical and grand canonical pictures

Consider a system of N particles whose physical states belong to the spaces $\mathfrak{h}_1, \dots, \mathfrak{h}_N$. The possible states of this system are described by the tensor product

$$\mathcal{H}_N = \mathfrak{h}_1 \otimes \cdots \otimes \mathfrak{h}_N.$$

Given the energy operators for each particle $h_1, \dots, h_N$, we lift these operators to the space $\mathcal{H}_N$ with the identification

$$h_i \longmapsto \mathbb{1} \otimes \cdots \otimes h_i \otimes \cdots \otimes \mathbb{1}.$$

The total energy of the system, given only the “internal” degrees of freedom for each particle and disregarding any interaction, is naturally given by

$$H_N^{\text{in}} = h_1 + \cdots + h_N \quad (2)$$

where

$$D(H_N^{\text{in}}) = \text{span}\{\phi_1 \otimes \cdots \otimes \phi_N : \phi_i \in D(h_i), i = 1, \dots, N\}$$

which is dense in $\mathcal{H}$ if $D(h_i)$ is dense in $\mathfrak{h}_i$ for all i .

Theorem 1.8 (ground state of non-interacting (distinguishable) particles). *If the ground state energies of the Hamiltonians $h_1, \dots, h_N$ are*

$$e_j = \inf_{\phi \in D(h_j), \|\phi\|=1} \langle \phi, h_j \phi \rangle, \quad j = 1, \dots, N$$

then the ground state energy of H_N^{in} is $\sum_{j=1}^N e_j$. Moreover, if $\phi_1, \dots, \phi_N$ are the ground states of these same Hamiltonians, then the ground state of H_N^{in} is $\phi_1 \otimes \cdots \otimes \phi_N$.

Proof. write proof. □

If we consider interactions we need to lift additional operators. For pair-wise interactions, although this procedure is fully generalizeable, to each pair (i, j) of distinct particles ($i \neq j$) we associate an interaction operator W_{ij} on a suitable domain contained in $\mathfrak{h}_i \otimes \mathfrak{h}_j$. To lift this operator to $\mathcal{H}_N$ we again make use of the identification

$$W_{ij} \longmapsto \mathbb{1} \otimes \cdots \otimes W_{ij} \otimes \cdots \otimes \mathbb{1},$$

which gives a tensor product of $N - 2$ identity operators and one interaction operator. Without specifying the correct domain we arrive at the formal way of writing the Hamiltonian of a system of N pair-wise interacting particles:

$$H_N = H_N^{\text{in}} + \sum_{1 \leq i < j \leq N} W_{ij} = \sum_{j=1}^N h_j + \sum_{1 \leq i < j \leq N} W_{ij}.$$

Requiring that the particles be indistinguishable from one another means that both their internal energy operators and the pair-wise interactions do not depend on the specific particle considered. This means that

$$\begin{aligned} \mathfrak{h}_1 &= \dots = \mathfrak{h}_N =: \mathfrak{h}; \\ h_1 &= \dots = h_N =: h; \\ W_{ij} &= W_{kl} =: W \quad \forall i \neq j, k \neq l. \end{aligned}$$

When lifting these operators to $\mathcal{H}_N$ we still need to specify indices that identify the specific factor of the tensor product they act on, for example

$$h_1 = h \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1}, \quad W_{12} = W \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1}.$$

With these restrictions, it is clear that H_N maps the spaces $\bigotimes_{\text{sym}}^N \mathfrak{h}$ and $\bigwedge^N \mathfrak{h}$ into themselves, which means that we can restrict this N-body Hamiltonian to either of the two subspaces and get a well defined operator (provided that a suitable domain for the starting Hamiltonian is chosen). (Consider putting the procedure in Forte's book to justify fermionic and bosonic statistics starting from exchange operator). Because the two subspaces mentioned are, respectively, the spaces of states that are symmetric and anti-symmetric under the exchange of any pair of particles, we have that the operators

$$P_+(H_N) \quad \text{and} \quad P_-(H_N)$$

are respectively the *bosonic* and *fermionic* N-body Hamiltonians for interacting particles.

This way of defining the energy of the system characterises the *canonical picture*, where the number of particles is fixed. But in statistical mechanics, allowing the number of particles to vary actually makes things easier, and as a byproduct also forces the introduction of an incredibly powerful formalism to describe quantum systems: second quantization. Letting N vary gets us to the *grand canonical picture*.

We start by defining a space that allows for superpositions of states with different numbers of particles, the *Fock space*. Given, for every N , the state spaces of N particles $\mathfrak{h}_1, \dots, \mathfrak{h}_N$, the Fock space is defined as

$$\mathcal{F} := \bigoplus_{N=0}^{\infty} \bigotimes_{i=1}^N \mathfrak{h}_i$$

where, as a convention, we interpret $\mathfrak{h}_1 \otimes \cdots \otimes \mathfrak{h}_N$ for $N = 0$ to be just $\mathbb{C}$. The state $1 \in \mathbb{C}$ is called the *vacuum state* and often denoted $|0\rangle$ or $|\Omega\rangle$. On this space, the energy operator is naturally given by

$$H = \bigoplus_{N=0}^{\infty} H_N, \quad \text{with} \quad H \bigoplus_{N=0}^{\infty} \Psi_N = \bigoplus_{N=0}^{\infty} H_N \Psi_N,$$

with H_0 taken to be the null operator (which means we are viewing the vacuum as having zero energy), and its domain is

$$D(H) = \left\{ \Psi = \bigoplus_{N=0}^{\infty} \Psi_N \in \mathcal{F} : \Psi_N \in D(H_N), \sum_{N=0}^{\infty} \|H_N \Psi_N\|^2 < \infty \right\}.$$

Introducing indistinguishability in this picture means repeating the steps we took for the N-body case in each *N-particle sector*, which gets us to the definition of the *fermionic* and *bosonic* Fock spaces. Given $\mathfrak{h}$ the state space for a single particle,

$$\begin{aligned} \mathcal{F}^B(\mathfrak{h}) &:= \bigoplus_{N=0}^{\infty} \bigotimes_{\text{sym}}^N \mathfrak{h} \\ \mathcal{F}^F(\mathfrak{h}) &:= \bigoplus_{N=0}^{\infty} \bigwedge^N \mathfrak{h} \end{aligned}$$

One of the most powerful aspects of the grand canonical picture is that all observables of the system can be written in the same way using special operators called *creation/annihilation operators*.

Starting from the Fock space $\mathcal{F} = \bigoplus_{N=0}^{\infty} \bigotimes^N \mathfrak{h}$ we define operators that add or remove a single particle state $f \in \mathfrak{h}$ as

$$\begin{aligned} a(f)(f_1 \otimes \cdots \otimes f_N) &:= \sqrt{N} \langle f, f_1 \rangle_{\mathfrak{h}} f_2 \otimes \cdots \otimes f_N, \\ a^*(f)(f_1 \otimes \cdots \otimes f_N) &:= \sqrt{N+1} f \otimes f_1 \otimes \cdots \otimes f_N. \end{aligned} \quad (3)$$

Here, for every $f \in \mathfrak{h}$, $a(f)$ is called *annihilation operator* and $a^*(f)$ is called *creation operator*. Note the very important special case where an annihilation operator acts on the vacuum: $a(f)|0\rangle = 0$ for all $f \in \mathcal{F}$, i.e. annihilation operators annihilate the vacuum. To indicate an operator that could be either of the two, we write $a^{\#}(f)$. With this definition the domain would be the set of pure tensors, but by linearity we can extend this to the space $\bigcup_{M=0}^{\infty} \bigoplus_{N=0}^M \mathcal{H}_N$, which is dense in $\mathcal{F}$.

Proposition 1.9. *For all $f \in \mathfrak{h}$, $a(f)$ and $a^*(f)$ can be extended to the domain*

$$\left\{ \Psi = \bigoplus_{N=0}^{\infty} \Psi_N : \sum_{N=0}^{\infty} N \|\Psi_N\|^2 < \infty \right\}.$$

Proof. ???

□

Proposition 1.10. *Creation and annihilation are **formally** the adjoint of one another, i.e. for all $f \in \mathfrak{h}$ and for all $\Psi, \Phi \in \bigcup_{M=0}^{\infty} \bigoplus_{N=0}^M \mathcal{H}_N$ it holds that*

$$\langle a(f)\Psi, \Phi \rangle_{\mathcal{F}} = \langle \Psi, a^*(f)\Phi \rangle_{\mathcal{F}}.$$

Proof. ???

□

Once again we introduce indistinguishability into the picture. Looking at the definition of these operators, it is clear that if a state $\Psi_N \in D(a^{\#}(f))$ is completely symmetric or completely anti-symmetric then applying an annihilation operator produces a state that is still completely symmetric or anti-symmetric (as Ψ_N can be written as a linear combination of all $N!$ permutations of N states and thus $a(f)$ removes all of these states exactly once), but applying a creation operator does introduce a state that is not yet accounted for in the (anti-)symmetrization. We then define *bosonic* (+) and *fermionic* (-) creation/annihilation operators as the projection onto the correct subspaces:

$$a_{\pm}(f) := a(f), \quad a_{\pm}^*(f) := P_{\pm}(a^*(f)) \quad \forall f \in \mathfrak{h}.$$

Remark 8. Bosonic and fermionic creation/annihilation operators are still densely defined and still formally adjoint to one another (within their respective statistics).

Remark 9. The map $\mathfrak{h} \ni f \mapsto a_{\pm}^*(f)$ is linear but $\mathfrak{h} \ni f \mapsto a_{\pm}(f)$ is conjugate linear.

Recall the definition of the commutator and anticommutator between two operators defined on a common domain: $\{A, B\} := AB + BA$ and $[A, B] := AB - BA$.

Proposition 1.11. *The operators a_{-} and a_{-}^* satisfy the Canonical Commutation Relations (CCR), while the operators a_{+} and a_{+}^* satisfy the Canonical Anticommutation Relations (CAR), i.e. for all $f, g \in \mathfrak{h}$ it holds that*

$$\begin{aligned} \{a_{+}(f), a_{+}^*(g)\} &= \langle f, g \rangle_{\mathfrak{h}}, & \{a_{+}(f), a_{+}(g)\} &= \{a_{+}^*(f), a_{+}^*(g)\} = 0 \\ [a_{-}(f), a_{-}^*(g)] &= \langle f, g \rangle_{\mathfrak{h}}, & [a_{-}(f), a_{-}(g)] &= [a_{-}^*(f), a_{-}^*(g)] = 0 \end{aligned}$$

Proof. ???

□

An important quantity we will come back to shortly is the number density operator. Given $u \in \mathfrak{h}$ a unit vector, $\bigotimes^N u$ is an eigenvector with eigenvalue N for the operator $a^*(u)a(u)$, which means this operator “counts” how many times a single-particle state is present in a state in $\mathcal{F}$. This is useful when considering the occupation number basis for a bosonic or fermionic Fock space, with the important distinction that bosonic modes can have arbitrary occupation number, while fermionic modes can only have occupation 0 or 1. Pauli’s exclusion principle is realized in quantum mechanics exactly by the antisymmetrization of fermionic states, indeed given two one-particle states $\phi, \psi \in \mathcal{H} \setminus \{0\}$, their antisymmetrization $\phi \otimes \psi - \psi \otimes \phi$ is not null if, and only if, $\phi \neq \psi$.

Proposition 1.12. *Let $u_1, \dots, u_r \in \mathfrak{h}$ be unit vectors and $\mathcal{H}_{n_1, \dots, n_r}$ the joint eigenspaces of the operators $a_\pm^*(u_1)a_\pm(u_1), \dots, a_\pm^*(u_r)a_\pm(u_r)$, then the fermionic and bosonic Fock spaces can be decomposed as direct sums of their joint eigenspaces. Indeed:*

$$\begin{aligned}\mathcal{F}^B(\mathfrak{h}) &= \bigoplus_{n_1=0}^{\infty} \cdots \bigoplus_{n_r=0}^{\infty} \mathcal{H}_{n_1, \dots, n_r} \\ \mathcal{F}^F(\mathfrak{h}) &= \bigoplus_{n_1=0}^1 \cdots \bigoplus_{n_r=0}^1 \mathcal{H}_{n_1, \dots, n_r}.\end{aligned}$$

Proof. We’ll prove the fermionic case, with $\dim(\mathfrak{h}) = M < \infty$ and $r = M$, as this provides the useful fermionic occupation number basis, which will be of great help to us later. Given $\{\phi_1, \dots, \phi_M\}$ an orthonormal basis for $\mathfrak{h}$, and calling $N_j = a_-^*(\phi_j)a_-(\phi_j) = a_j^*a_j$, we can prove that these number operators are commuting projections. These operators are indeed self-adjoint, as

$$N_j^* = (a_j^*a_j)^* = a_j^*a_j = N_j,$$

and idempotent, as the CAR imply

$$N_j^2 = (a_j^*a_j)^2 = -a_j^*a_j^*a_ja_j + a_j^*a_j = a_j^*a_j.$$

Moreover, for $j \neq k$, we have that

$$[N_j, N_k] = a_j^*a_ja_k^*a_k - a_k^*a_ka_j^*a_j = a_k^*a_ka_j^*a_j - a_k^*a_ka_j^*a_j = 0,$$

therefore, all the number operators are projections and can be all diagonalized simultaneously. Note that these algebraic manipulations are well defined, since all the operators $a^\#$ (and their products) are defined on the entire space $\mathcal{F}^F(\mathfrak{h})$, which is finite-dimensional if $\mathfrak{h}$ is finite-dimensional. We now turn to the direct sum decomposition. Let $\mathbf{n} = (n_1, \dots, n_M) \in \{0, 1\}^M$ be an M -tuple of eigenvalues of the respective number operators, and define the joint projection as

$$P_{\mathbf{n}} := \prod_{i=1}^M (n_i N_i + (1 - n_i)(\mathbb{1} - N_i)).$$

This is a product of commuting projections and is therefore still a projection, and still defined on the entire fermionic Fock space. The set $\{P_{\mathbf{n}} : \mathbf{n} \in \{0, 1\}^M\}$ is a complete set of projections, as shown by the completeness relation

$$\begin{aligned}\sum_{\mathbf{n} \in \{0, 1\}^M} P_{\mathbf{n}} &= \sum_{\mathbf{n}} \prod_{i=1}^M (n_i N_i + (1 - n_i)(\mathbb{1} - N_i)) \\ &= \prod_{i=1}^M \sum_{n_i=0}^1 (n_i N_i + (1 - n_i)(\mathbb{1} - N_i)) \\ &= \prod_{i=1}^M \mathbb{1} = \mathbb{1},\end{aligned}$$

again defined on the entire Fock space. We conclude by simply applying this completeness relation:

$$\mathcal{F}^F(\mathfrak{h}) = \sum_{\mathbf{n} \in \{0,1\}^M} P_{\mathbf{n}} \mathcal{F}^F(\mathfrak{h}) = \bigoplus_{\mathbf{n} \in \{0,1\}^M} P_{\mathbf{n}} \mathcal{F}^F(\mathfrak{h}).$$

□

Remark 10. The case where $\mathfrak{h}$ is infinite-dimensional is not much different, and even then we can always choose a finite set of M independent number operators to obtain a decomposition into a direct sum of M (possibly infinite-dimensional) joint eigenspaces. But, as will be made clear later on, we will put a UV cutoff on our system and this will reduce the allowed momenta of a particle to a finite set. Having then a finite one-particle state space will make our life much easier, as we can make use of the finite basis given by the eigenvectors $\{|\mathbf{n}\rangle = |n_1\rangle \otimes \cdots \otimes |n_M\rangle : n_i \in \{0,1\} \forall i = 1, \dots, M\}$, with eigenvalues given by the occupation of the M independent modes.

We can now come back to a statement that we previously used as motivation for the introduction of creation/annihilation operators, and show that indeed every observable in the grand canonical picture can be written in terms of such operators. We focus only on 1-body and 2-body observables because these are the types of operators that show up in the Hamiltonian we set out to study, but this procedure can be generalized to any N -body operator.

Theorem 1.13 (2nd quantization of 1-body operator). *Let h be an observable on $\mathfrak{h}$ and $\{u_\alpha\}_{\alpha=1}^\infty$ be an orthonormal basis for $\mathfrak{h}$ with elements in the dense domain $D(h)$. Following the notation from the beginning of this section, we can write*

$$\bigoplus_{N=1}^\infty \sum_{j=1}^N h_j = \sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle_{\mathfrak{h}} a_\pm^*(u_\alpha) a_\pm(u_\beta)$$

on the domain $\cup_{M=0}^\infty \bigoplus_{N=1}^M P_\pm(D(\sum_{j=1}^N h_j))$ (for $M = 0$ the domain is taken to be $\mathbb{C}$). This operator is called the *second quantization* of h .

Proof. First we observe that the thesis holds in the sense of quadratic forms on $D(h)$. Indeed, let $f, g \in D(h)$ and recall the definition of creation/annihilation operators in 3

$$\begin{aligned} \sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle \langle g, a_\pm^*(u_\alpha) a_\pm(u_\beta) f \rangle &= \sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle \langle g, u_\alpha \rangle \langle u_\beta, f \rangle = \sum_{\beta=1}^\infty \langle g, h u_\beta \rangle \langle u_\beta, f \rangle \\ &= \sum_{\beta=1}^\infty \langle h g, u_\beta \rangle \langle u_\beta, f \rangle = \langle h g, f \rangle = \langle g, h f \rangle \end{aligned}$$

where the indices α and β are removed by using the completeness of the basis $\{u_\alpha\}_{\alpha=1}^\infty$, and h can be moved on either side of the inner products as it is a symmetric operator with $f, g \in D(h)$ and $u_\alpha \in D(h)$ for all α . Moving onto N -fold pure tensors, let $f_1, \dots, f_N \in D(h)$ for some $N \geq 1$ and consider the action of $a^*(u_\alpha) a(u_\beta)$ on their tensor product

$$a^*(u_\alpha) a(u_\beta) f_1 \otimes \cdots \otimes f_N = N \langle u_\beta, f_1 \rangle u_\alpha \otimes f_2 \otimes \cdots \otimes f_N$$

then it follows that

$$\sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle a^*(u_\alpha) a(u_\beta) = N h_1$$

as quadratic form on finite linear combinations of pure tensor products of elements in $D(h)$. Recall the definition of the projections $P_\pm$ from equation 1, then it is clear that

$$P_\pm N h_1 P_\pm = P_\pm \sum_{j=1}^N h_j P_\pm = \sum_{j=1}^N h_j P_\pm$$

on $\bigotimes^N \mathfrak{h}$, as each h_j acts in the same way as h on the respective factor in the tensor product, which means that it maps the (anti-)symmetric subspace into itself. Then it follows that

$$\sum_{\alpha, \beta=1}^{\infty} \langle u_\alpha, h u_\beta \rangle a_\pm^*(u_\alpha) a_\pm(u_\beta) P_\pm = \sum_{\alpha, \beta=1}^{\infty} \langle u_\alpha, h u_\beta \rangle P_\pm a^*(u_\alpha) a(u_\beta) P_\pm = \bigoplus_{N=1}^{\infty} \sum_{j=1}^N h_j P_\pm$$

from which the thesis follows immediately. $\square$

Remark 11. Another way to lift an operator U to the Fock space is multiplicatively as $\bigoplus_{N=0}^{\infty} \bigotimes^N U$, and this plays an important role in the discussion of symmetries. Given a Lie group (although this discussion can be somewhat generalized to groups that are not Lie groups) and a unitary representation γ , then $d\gamma$ defines a representation of its associated Lie algebra as hermitian operators. One can then define a map Γ that lifts the representation γ multiplicatively to the Fock space, and this induces again another representation $d\Gamma$ which now lifts the representation $d\gamma$ additively to the Fock space. One can show that Γ indeed defines a representation, and this highlights how the two ways of lifting an operator to a Fock space respect two different algebraic structures (at least in this case where we discuss groups of symmetries of a system): a multiplicative (group) and a linear (algebra) structure. Given a unitary operator U on $\mathfrak{h}$ we can call $\Gamma(U)$ its multiplicative representation on the Fock space, and given a symmetric operator A we can call $d\Gamma(A)$ its additive representation on the Fock space.

Remark 12. The second quantization of the identity operator on $\mathfrak{h}$ is the number operator

$$\mathcal{N} = \bigoplus_{N=0}^{\infty} \sum_{j=0}^N \mathbb{1}_j = \bigoplus_{N=0}^{\infty} N = \sum_{\alpha=1}^{\infty} a_\pm^*(u_\alpha) a_\pm(u_\beta).$$

Now it is clear why we called $a_\pm^*(f) a_\pm(f)$ the number *density* operator, as a number operator is the sum of operators that count the occupation number of a given state over a “complete set” of states. In a much more delicate way one could define creation/annihilation *distributions* as objects that formally create/annihilate states completely localized at one point in space. If we called $|x\rangle$ an “eigenstate” (in a generalized sense that we will not discuss here) of the position operator with eigenvalue x , then the number density would be written as $\rho(x) = a_\pm^*(|x\rangle) a_\pm(|x\rangle)$ and the total number of particles would turn into a very familiar expression: $\mathcal{N} = \int_{\mathbb{R}^n} dx \rho(x)$, but to make sense of these objects we would need tools that go beyond the scope of our work.

Theorem 1.14 (2nd quantization of 2-body operator). *Let W be a 2-body potential for identical particles, i.e. an observable on $\mathfrak{h} \otimes \mathfrak{h}$ such that (you didn't cover this in the section on tensor products) $\text{Ex}W\text{Ex} = W$. Let $\{u_\alpha\}_{\alpha=1}^{\infty}$ be an orthonormal basis for $\mathfrak{h}$ such that $u_\alpha \otimes u_\beta \in D(W)$ for all α and β . Then as a quadratic form on the domain $\cup_{M=2}^{\infty} \bigoplus_{N=2}^M P_\pm(D(\sum_{1 \leq i < j \leq N} W_{ij}))$ we have*

$$\bigoplus_{N=2}^{\infty} \sum_{1 \leq i < j \leq N} W_{ij} = \frac{1}{2} \sum_{\alpha, \beta, \nu, \mu=1}^{\infty} \langle u_\alpha \otimes u_\beta, W u_\mu \otimes u_\nu \rangle_{\mathfrak{h} \otimes \mathfrak{h}} a_\pm^*(u_\alpha) a_\pm^*(u_\beta) a_\pm(u_\nu) a_\pm(u_\mu).$$

Again, this operator is called the second quantization of W . Note that in the fermionic case, the order in which the indices appear in the creation-creation and annihilation-annihilation products matters, as these operators anticommute rather than commute.

Proof. prooooooooo $\square$

1.3 One-particle density matrices

It is clear from our discussion of the second quantization of an operator, that expectation values of any observable will inevitably reduce to expectation values of products of creation and annihilation operators. An important result that we will discuss in the section after this one states that – under appropriate conditions which are not at all obvious – the expectation of an observable can be written in terms of expectations of operators of the form $a_{\pm}^{\#} a_{\pm}^{\#}$. We therefore begin the study of these kinds of expectations, starting from the following definition.

Definition 1.16 (One-particle density matrix (1-pdm)). Let $\Psi = \bigoplus_{N=0}^{\infty} \Psi_N$ be a normalized vector on the bosonic or fermionic Fock space $\mathcal{F}^{\text{B,F}}(\mathfrak{h})$ with finite particle number expectation

$$\langle \Psi, \mathcal{N} \Psi \rangle = \sum_{N=0}^{\infty} N \|\Psi_N\|^2 < \infty.$$

We define the *1-particle density matrix* or *2-point function* of Ψ as the operator γ_{Ψ} on $\mathfrak{h}$ given, as a quadratic form, by

$$\langle f, \gamma_{\Psi} g \rangle_{\mathfrak{h}} = \langle \Psi, a_{\pm}^*(g) a_{\pm}(f) \Psi \rangle$$

for all $f, g \in \mathfrak{h}$.

Proposition 1.15. *The operator γ_{Ψ} is positive semi-definite (btw you didn't introduce partial ordering of operators) and trace class, with*

$$\text{Tr}(\gamma_{\Psi}) = \langle \Psi, \mathcal{N} \Psi \rangle.$$

Proof. ??? □

Lemma 1.16. *If Ψ is a finite linear combination of pure tensor products of elements from a subspace $X \subseteq \mathfrak{h}$ then γ_{Ψ} is a finite rank operator whose range is a subspace of X .*

Proof. ??? □

Proposition 1.17. *If $\Psi \in \mathcal{F}^{\text{F}}(\mathfrak{h})$ is a normalized vector then γ_{Ψ} satisfies the inequality*

$$0 \leq \gamma_{\Psi} \leq \mathbb{1}.$$

In particular, the eigenvalues of γ_{Ψ} lie in the interval $[0, 1]$.

Proof. We need to show that for every $f \in \mathfrak{h}$ it holds that $0 \leq \langle f, \gamma_{\Psi} f \rangle \leq \|f\|^2$. Using that $a(f)$ and $a^*(f)$ are formally adjoints of each other, we can see that

$$\langle f, \gamma_{\Psi} f \rangle = \langle \Psi, a_{-}^*(f) a_{-}(f) \Psi \rangle = \langle a_{-}(f) \Psi, a_{-}(f) \Psi \rangle = \|a_{-}(f) \Psi\|^2 \geq 0.$$

On the other hand, the CAR imply

$$\begin{aligned} \langle f, \gamma_{\Psi} f \rangle &= \langle \Psi, a_{-}^*(f) a_{-}(f) \Psi \rangle \leq \langle \Psi, a_{-}^*(f) a_{-}(f) \Psi \rangle + \langle a_{-}^*(f) \Psi, a_{-}^*(f) \Psi \rangle \\ &= \langle \Psi, (a_{-}^*(f) a_{-}(f) + a_{-}(f) a_{-}^*(f)) \Psi \rangle = \langle \Psi, \{a_{-}^*(f), a_{-}(f)\} \Psi \rangle = \|f\|^2. \end{aligned}$$

This concludes the proof. □

We are now ready to prove a fundamental theorem on which the treatment of any fermionic system is built, especially if perturbation theory is involved: the ground state energy and eigenvector of non-interacting fermions.

Theorem 1.18 (Ground state of non-interacting fermions). *Consider the Hamiltonian $H_N^{\text{in}} = \sum_{j=1}^N h_j$ for N particles, introduced in Equation 2, restricted to the anti-symmetric subspace $\bigwedge^N \mathfrak{h}$, i.e. with domain*

$$D_-(H_N^{\text{in}}) = P_- D(H_N^{\text{in}}).$$

Assume that the one-body operator h is bounded below. The ground state energy of this fermionic system is

$$\inf \{ \text{Tr}(Ph) : P \text{ is a projection onto an } N\text{-dimensional subspace of } D(h) \}.$$

If the infimum is attained for an N -dimensional projection P then H_N^{in} has a ground state eigenvector $f_1 \wedge \dots \wedge f_N$, where $f_1, \dots, f_N$ is an orthonormal basis for $P(\mathfrak{h})$. In particular, these vectors can be chosen to be eigenvectors of h , and the expectations of h restricted to $P(\mathfrak{h})^\perp \cap D(h)$ will be greater than the expectations of h restricted to $P(\mathfrak{h})$.

Proof. Since $D(h)$ is assumed to be dense, we can choose an orthonormal basis $\{u_\alpha\}_{\alpha=1}^\infty$ for $\mathfrak{h}$ of elements in $D(h)$. Let $\Psi \in D_-(H_N^{\text{in}})$ be a normalized vector, then

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle = \sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle \langle \Psi, a_-^*(u_\alpha) a_-(u_\beta) \Psi \rangle = \sum_{\alpha, \beta=1}^\infty h_{\alpha\beta} \langle u_\beta, \gamma_\Psi u_\alpha \rangle.$$

Since Ψ is assumed to be a finite linear combination of pure tensor products of elements of $D(h)$, it follows from Lemma 1.16 that γ_Ψ has finite rank. Let $\{v_1, \dots, v_n\}$ be the set of eigenvectors of γ_Ψ with eigenvalues $\{\lambda_1, \dots, \lambda_n\}$, then it follows from the same lemma that $v_1, \dots, v_n \in D(h)$. Then

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle = \sum_{j=1}^n \sum_{\alpha, \beta=1}^\infty \lambda_j h_{\alpha\beta} \langle u_\beta, v_j \rangle \langle v_j, u_\alpha \rangle = \sum_{j=1}^n \lambda_j \langle v_j, h v_j \rangle.$$

Because $\Psi \in D_-(H_N^{\text{in}})$, it must have fixed particle number N , then it follows that

$$\sum_{j=1}^n \lambda_j = \text{Tr}(\gamma_\Psi) = \langle \Psi, \mathcal{N} \Psi \rangle = N,$$

but $0 \leq \lambda_j \leq 1$, and therefore it must be that $n \geq N$. We may assume the eigenvectors v_j to be ordered such that $\langle v_1, h v_1 \rangle \leq \dots \leq \langle v_n, h v_n \rangle$ so that, by defining P as the projection onto $\text{span}\{v_1, \dots, v_N\}$, we can provide the bound

$$\text{Tr}(Ph) = \sum_{j=1}^N \langle v_j, h v_j \rangle \leq \sum_{j=1}^n \lambda_j \langle v_j, h v_j \rangle = \langle \Psi, H_N^{\text{in}} \Psi \rangle.$$

Thus

$$\begin{aligned} & \inf \{ \text{Tr}(Ph) : P \text{ proj. onto } N\text{-dimensional subspace of } D(h) \} \\ & \leq \inf \{ \langle \Psi, H_N^{\text{in}} \Psi \rangle : \Psi \in D(H_N^{\text{in}}), \|\Psi\| = 1 \}. \end{aligned}$$

It remains to show the opposite inequality. Let $\Psi = f_1 \wedge \dots \wedge f_N$ with $f_1, \dots, f_N \in D(h)$ orthonormal vectors. Again $\gamma_\Psi =: P$ is a projection onto $\text{span}\{f_1, \dots, f_N\}$, and as above we find that

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle \leq \text{Tr}(\gamma_\Psi h)$$

and the inequality carries to the infimum, which finally proves the double inequality and the first part of the thesis.

Assume now that P is a minimizing projection for the variational problem of the thesis.

AAAAAHHHH.

□

Before going over the next section there is an important generalization to be made. Everything up to now relied on the possibility of writing the second quantization of any observable by means of products of creation and annihilation operators, once a basis for the one-particle Hilbert space was chosen. It is, however, quite natural to ask what happens - among other things - when a different basis is chosen, or when the Hamiltonian contains terms of the form $a^*(f)a^*(g)$ and $a(f)a(g)$. These, and other questions, can be answered by appropriately generalizing the one-particle density matrix, and the aim of this section is precisely to introduce this generalization. Our study case doesn't include terms of the above form, i.e. it only contains number-preserving interactions, but when going over Wick's theorem it will become quite clear just how powerful this generalization truly is, as it would eventually make it much easier to study the same problem through different physical lenses, using quasi-particle states, and also study much more general systems.

Definition 1.17 (Generalized 1pdm). Given a normalized vector $\Psi \in \mathcal{F}^{\text{F,B}}(\mathfrak{h})$ with finite particle number expectation, i.e. $\langle \Psi, \mathcal{N}\Psi \rangle < \infty$, we define the *generalized one-particle density matrix* as the positive semi-definite operator $\Gamma_\Psi : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*$ defined by

$$\langle f_1 \oplus Jg_1, \Gamma_\Psi f_2 \oplus Jg_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*} = \langle \Psi, (a_\pm^*(f_2) + a_\pm(g_2))(a_\pm(f_1) + a_\pm^*(g_1))\Psi \rangle_{\mathcal{F}^{\text{F,B}}}$$

where J is the canonical isomorphism between $\mathfrak{h}$ and $\mathfrak{h}^*$.

It will be useful to recall that J acts as $Jg(f) = \langle g, f \rangle$, and its inverse is also its adjoint, i.e. $J^{-1} = J^*$. Given the normalized vector Ψ we may write Γ_Ψ in block-matrix form by using an auxiliary operator, $\alpha_\Psi : \mathfrak{h}^* \rightarrow \mathfrak{h}$, that “measures” how far the state Ψ is from having a fixed particle number. The operator is defined by

$$\langle f, \alpha_\Psi Jg \rangle = \langle \Psi, a_\pm(g)a_\pm(f)\Psi \rangle$$

for all $f, g \in \mathfrak{h}$. Note that this map has a useful property, which essentially relates its adjoint to the corresponding operator $a_\pm^*(f)a_\pm^*(g)$, that reads

$$\alpha_\Psi^* = \pm J\alpha_\Psi J,$$

as can be shown by noting that, using the CAR or the CCR,

$$\langle \alpha_\Psi^* f, Jg \rangle = \langle f, \alpha_\Psi Jg \rangle = \pm \langle g, \alpha_\Psi Jf \rangle = \pm \langle J\alpha_\Psi Jf, Jg \rangle.$$

With this operator we come to the following lemma.

Lemma 1.19. *Given a normalized state $\Psi \in \mathcal{F}^{\text{F,B}}(\mathfrak{h})$ with finite particle number expectation, its generalized one-particle density matrix can be written in the block-matrix form as*

$$\Gamma_\Psi = \begin{pmatrix} \gamma_\Psi & \alpha_\Psi \\ \pm J\alpha_\Psi J & \mathbb{1} \pm J\gamma_\Psi J^* \end{pmatrix}$$

Proof. It suffices to prove that, as a quadratic form on $\mathfrak{h} \oplus \mathfrak{h}^*$, this matrix reproduces exactly Definition 1.17. Let $f_1, g_1, f_2, g_2 \in \mathfrak{h}$, then it holds that

$$\begin{aligned} (f_1 \quad Jg_1) \begin{pmatrix} \gamma_\Psi & \alpha_\Psi \\ \pm J\alpha_\Psi J & \mathbb{1} \pm J\gamma_\Psi J^* \end{pmatrix} \begin{pmatrix} f_2 \\ Jg_2 \end{pmatrix} &= (f_1 \quad Jg_1) \begin{pmatrix} \gamma_\Psi f_2 + \alpha_\Psi Jg_2 \\ \pm J\alpha_\Psi Jf_2 + (\mathbb{1} \pm J\gamma_\Psi J^*)Jg_2 \end{pmatrix} \\ &= \langle f_1, \gamma_\Psi f_2 \rangle + \langle f_1, \alpha_\Psi Jg_2 \rangle + \langle Jg_1, \alpha_\Psi^* f_2 \rangle + \langle Jg_1, (\mathbb{1} \pm J\gamma_\Psi J^*)Jg_2 \rangle \\ &= \langle a_\pm^*(f_2)a_\pm(f_1) \rangle + \langle a_\pm(g_2)a_\pm(f_1) \rangle + \langle a_\pm^*(f_2)a_\pm^*(g_1) \rangle + \langle a_\pm(g_2)a_\pm^*(g_1) \rangle \\ &= \langle (a_\pm^*(f_2) + a_\pm(g_2))(a_\pm(f_1) + a_\pm^*(g_1)) \rangle \\ &= \langle f_1 \oplus Jg_1, \Gamma_\Psi f_2 \oplus Jg_2 \rangle. \end{aligned}$$

□

The usefulness of this new object is that it allows for a very simple way to write this quantity in a way that closely resembles the standard one-particle density matrix. We just need to define the generalized creation/annihilation operators:

Definition 1.18. Let $f, g \in \mathfrak{h}$ be one-particle states. The generalized annihilation and creation operators are defined, respectively, by the maps

$$\begin{aligned} A_{\pm}(f \oplus Jg) &:= a_{\pm}(f) + a_{\pm}^*(g) \\ A_{\pm}^*(f \oplus Jg) &:= a_{\pm}^*(f) + a_{\pm}(g). \end{aligned}$$

Remark 13. Once again we have that the generalized annihilation operators are conjugate linear from $\mathfrak{h} \oplus \mathfrak{h}^*$ to operators on $\mathcal{F}^{F,B}(\mathfrak{h})$. Furthermore, we have the following relation between these generalized operators. For all $F \in \mathfrak{h} \oplus \mathfrak{h}^*$ it holds that

$$A_{\pm}^*(F) = A_{\pm}(\mathcal{J}F), \quad \text{where } \mathcal{J} = \begin{pmatrix} 0 & J^* \\ J & 0 \end{pmatrix} : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*$$

Proposition 1.20 (Generalized CCR and CAR). *For all $F_1, F_2 \in \mathfrak{h} \oplus \mathfrak{h}^*$ we have the following generalized versions of the canonical (anti-)commutation relations. For bosons*

$$[A_+(F_1), A_+^*(F_2)] = \langle F_1, \mathcal{S}F_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*} \quad \text{where } \mathcal{S} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix} : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*,$$

and for fermions

$$\{A_-(F_1), A_-^*(F_2)\} = \langle F_1, F_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*}.$$

More compactly we could introduce the natural notation $[A_{\pm}(F_1), A_{\pm}^(F_2)]_{\mp} = \langle F_1, \mathcal{S}_{\pm}F_2 \rangle$, where we define $\mathcal{S}_{-}$ as simply being the identity, and the sign on the square brackets indicates a commutator $(-)$ or an anticommutator $(+)$.*

Proof. **IT'S JUST EXPLICIT CALCULATION.** □

Remark 14. Note that, in general, the (anti-)commutators $[A_{\pm}(F), A_{\pm}(G)]_{\mp}$ and $[A_{\pm}^*(F), A_{\pm}^*(G)]_{\mp}$ need not be zero.

Remark 15. The special case where $F = f \oplus 0$ reads $A_{\pm}^{\#}(F) = a_{\pm}^{\#}(f)$.

With these operators, the generalized one-particle density matrix can be written in a familiar form. Given, as usual, $\Psi \in \mathcal{F}^{F,B}(\mathfrak{h})$ normalized and for all $F_1, F_2 \in \mathfrak{h} \oplus \mathfrak{h}^*$ we have

$$\langle F_1, \Gamma_{\Psi} F_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*} = \langle \Psi, A_{\pm}^*(F_1) A_{\pm}(F_2) \Psi \rangle$$

1.4 Bogolubov transformations and quasi-free pure states

We can finally realize the notion of a “change of basis” for creation/annihilation operators on the space $\mathfrak{h} \oplus \mathfrak{h}^*$. The following definition characterizes a class of isomorphisms on this space as those that, in some sense, preserve the structure of the generalized creation and annihilation operators.

Definition 1.19 (Bogolubov transformations). A linear bounded isomorphism $\mathcal{V} : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*$ is called a (bosonic or fermionic) *Bogolubov map* if, for all $F, G \in \mathfrak{h} \oplus \mathfrak{h}^*$

- $A^*(\mathcal{V}F) = A(\mathcal{V}JF)$;
- $[A_{\pm}(\mathcal{V}F), A_{\pm}^*(\mathcal{V}G)]_{\mp} = \langle F, \mathcal{S}_{\pm}G \rangle$.

One also refers to the map $A_{\pm}(F) \mapsto A_{\pm}(\mathcal{V}F)$ as a *Bogolubov transformation*.

We can characterize Bogolubov maps by writing them in block-matrix form and noting that each block must satisfy specific properties that come directly from the definition. We first notice that the conditions in the definition can be rewritten as

$$\langle \mathcal{V}F_1, \mathcal{S}_\pm \mathcal{V}F_2 \rangle = \langle F_1, \mathcal{S}_\pm F_2 \rangle \quad \text{and} \quad \mathcal{J}\mathcal{V}F = \mathcal{V}\mathcal{J}F$$

for all $F, F_1, F_2 \in \mathfrak{h} \oplus \mathfrak{h}^*$. Moreover, since we are assuming that $\mathcal{V}$ is invertible, we can also write

$$\mathcal{V}^{-1} = \mathcal{S}_\pm \mathcal{V}^* \mathcal{S}_\pm.$$

In particular, $\mathcal{V}^{-1} = \mathcal{V}^*$ in the fermionic case. With these properties we immediatly get to the following proposition, for which the proof can be read through what we just said:

Proposition 1.21. *A linear map $\mathcal{V}$ on $\mathfrak{h} \oplus \mathfrak{h}^*$ is a Bogolubov map if, and only if, the following conditions hold:*

$$\mathcal{V}^* \mathcal{S}_\pm \mathcal{V} = \mathcal{S}_\pm, \quad \mathcal{V} \mathcal{S}_\pm \mathcal{V}^* = \mathcal{S}_\pm, \quad \mathcal{J}\mathcal{V}\mathcal{J} = \mathcal{V}. \quad (4)$$

If we take bounded linear maps $U : \mathfrak{h} \rightarrow \mathfrak{h}$ and $V : \mathfrak{h} \rightarrow \mathfrak{h}^*$ we can finally write a Bogolubov map in block-matrix form

$$\mathcal{V} = \begin{pmatrix} U & J^* V J^* \\ V & J U J^* \end{pmatrix} \quad (5)$$

and the conditions 4 can then be read as

$$\begin{cases} U^* U = \mathbb{1} \pm V^* V \\ U U^* = \mathbb{1} \pm J^* V^* J \\ J V^* J U = \pm J U^* J^* V \end{cases}.$$

Defining Bogolubov maps is not sufficient though, as such a map need not, in general, correspond to a unitary on the Fock space. Thus we come to the following theorem which tells us when this is possible, and which we will use to define an important class of states in a Fock space for which Wick's theorem will hold in its most general form.

Theorem 1.22 (Unitary implementability of Bogolubov maps). *If $\mathcal{V} : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*$ is a Bogolubov map of the form 5 then there exists a unitary map*

$$\mathbb{U}_\mathcal{V} : \mathcal{F}^{\mathbb{F}, \mathbb{B}}(\mathfrak{h}) \rightarrow \mathcal{F}^{\mathbb{F}, \mathbb{B}}(\mathfrak{h})$$

such that, for $F \in \mathfrak{h} \oplus \mathfrak{h}^*$,

$$\mathbb{U}_\mathcal{V} A_\pm(F) \mathbb{U}_\mathcal{V}^* = A_\pm(\mathcal{V}F)$$

if, and only if, $V^* V$ is trace-class. This trace-class condition is also referred to as the Shale-Stinespring condition.

What this means is that, under suitable conditions on the block-matrix form of a Bogolubov map, there exists a “change of basis” on the Fock space that sends generalized annihilation and - by taking the adjoint - creation operators to operators that annihilate/create a superposition of states determined by the linear map $\mathcal{V}$.

Proof. AAAAAAHHHHHHH

□

We are finally ready to discuss the most important concept in this chapter: Wick's theorem. This is the result that, among other things, allows us to write perturbative series as they are described in the next chapter, and it is motivation behind the computational tools we will need to introduce. Thanks to the formalism of second quantization, Wick's theorem essentially states that the expectation value for any observable on a special class of states is completely determined by the 2-point function. Moreover, since the perturbative expansion of a physical quantity involves a series of products of creation/annihilation operators, this theorem also says that such a perturbative expansion is completely determined by the 2-point function as well.

Definition 1.20 (Quasi-free pure state). A vector $\Psi \in \mathcal{F}^{\text{F,B}}(\mathfrak{h})$ is called a *quasi-free pure state* if there exists a unitarily implementable Bogolubov map $\mathcal{V}$ such that $\Psi = \mathbb{U}_{\mathcal{V}}|0\rangle$, where $\mathbb{U}_{\mathcal{V}}$ is the unitary implementation of $\mathcal{V}$.

That is, a state is quasi-free if it is the vacuum state under an appropriate choice of physical modes.

Theorem 1.23 (Wick's theorem). *If Ψ is a quasi-free pure state and $F_1, \dots, F_{2m} \in \mathfrak{h} \oplus \mathfrak{h}^*$ for some $m \geq 1$, then it holds that*

$$\langle \Psi, A_{\pm}(F_1) \cdots A_{\pm}(F_{2m}) \Psi \rangle = \sum_{\sigma \in P_{2m}} (\pm 1)^{\sigma} \langle \Psi, A_{\pm}(F_{\sigma(1)}) A_{\pm}(F_{\sigma(2)}) \Psi \rangle \cdots \langle \Psi, A_{\pm}(F_{\sigma(2m-1)}) A_{\pm}(F_{\sigma(2m)}) \Psi \rangle$$

and

$$\langle \Psi, A_{\pm}(F_1) \cdots A_{\pm}(F_{2m-1}) \Psi \rangle = 0$$

where P_{2m} is the set of pairings of the indices $\{1, \dots, 2m\}$, i.e.

$$P_{2m} = \{ \sigma \in S_{2m} : \sigma(2j-1) < \sigma(2j+1), j = 1, \dots, m-1 \text{ and } \sigma(2j-1) < \sigma(2j), j = 1, \dots, m \}.$$

Despite dreaming of resisting the urge to characterize *something*, the mathematician is ultimately doomed to fail. Let us then indulge in the characterization of generalized 2-point functions of quasi-free pure states, since they appear to be so important.

Proposition 1.24. *If $\Psi \in \mathcal{F}^{\text{F,B}}(\mathfrak{h})$ is a quasi-free pure state then the generalized 2-point function, $\Gamma = \Gamma_{\Psi}$, satisfies the condition*

$$\Gamma \mathcal{S}_{\pm} \Gamma = \mp \Gamma. \quad (6)$$

In particular, Γ is a projection in the fermionic case. Conversely, if $\Gamma : \mathfrak{h} \oplus \mathfrak{h}^ \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*$ is a positive semi-definite operator satisfying 6 and of the form*

$$\Gamma = \begin{pmatrix} \gamma & \alpha \\ \pm J \alpha J & \mathbb{1} \pm J \gamma J^* \end{pmatrix}$$

with γ a trace-class operator, then there exists a quasi-free pure state $\Psi \in \mathcal{F}^{\text{F,B}}(\mathfrak{h})$ such that $\Gamma_{\Psi} = \Gamma$.

Proof. proof you have to write, if mathematician you want to become, mmh. □

1.5 Quadratic hamiltonians

Having introduced all the necessary results regarding Bogolubov maps, we can now move onto the study of generic quadratic Hamiltonians. Their importance is partly due to the fact that if one can

find a Bogolubov transformation that diagonalizes the quadratic part of the Hamiltonian, then the study of the kinetic (quadratic) part of that system is automatically done. This of course plays a crucial role in perturbative calculations, which are all about dealing with complicated interactions on top of an easy and well-defined “free theory”, i.e. a theory whose n -point functions satisfy Wick’s theorem.

Definition 1.21 (Quadratic Hamiltonian). Let $\mathcal{A} : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*$ be a self-adjoint, trace-class operator and assume moreover in the bosonic case that it is positive semi-definite. Given an orthonormal basis $\{F_1, \dots, F_{2M}\}$ for $\mathfrak{h} \oplus \mathfrak{h}^*$, the operator

$$H_{\mathcal{A}}^{\pm} = \sum_{i,j=1}^{2M} \langle F_i, \mathcal{A} F_j \rangle A_{\pm}^*(F_i) A_{\pm}(F_j),$$

is called the *quadratic Hamiltonian associated to $\mathcal{A}$* .

2 Perturbation Theory

2.1 A toy model

To compute expectation values of products of field operators, one can follow the Feynman integral approach. The aim is to define this integral without trying to integrate over field configurations, and in some cases this can be done rigorously by defining this special integral as a linear operator on commuting (bosonic) or anticommuting (fermionic) variables in such a way that the integral of polynomials respects Wick’s rule. Bosonic field operators can be represented with complex variables, this way their expectation values become complex Gaussian integrals. Fermionic field operators on the other hand require a different algebraic structure.

2.2 Bosonic expectations

2.3 Grassmann Algebras and Grassmann Integrals

Consider the unital complex Grassmann algebra generated by a set of $2N$ anticommuting vectors:

$$\begin{aligned} \mathcal{G}_N &= \text{span}\{1, \psi_1, \bar{\psi}_1, \dots, \psi_N, \bar{\psi}_N\} \\ \{\psi_i, \psi_j\} &= 0 \quad \forall i, j = 1, \dots, N \\ \{\psi_i, \bar{\psi}_j\} &= 0 \quad \forall i, j = 1, \dots, N \\ \{\bar{\psi}_i, \bar{\psi}_j\} &= 0 \quad \forall i, j = 1, \dots, N \end{aligned} \tag{7}$$

where $\{\psi, \phi\} := \psi\phi + \phi\psi$ is the anticommutator. The idea is to interpret the elements of this algebra as (polynomials of) field operators, so we need to define integrals and expectations of Grassmann variables, i.e. of elements of this algebra.

Given $\mathcal{G}_N$ the Grassmann algebra generated by $\{\psi_1, \dots, \psi_N, \bar{\psi}_1, \dots, \bar{\psi}_N\}$, denote by

$$\begin{aligned} \mathcal{G}_N^- &= \{\phi \in \mathcal{G}_N : [\phi, \psi] = 0 \quad \forall \psi \in \mathcal{G}_N\} \\ \mathcal{G}_N^+ &= \{\phi \in \mathcal{G}_N : \{\phi, \psi\} = 0 \quad \forall \psi \in \mathcal{G}_N\} \end{aligned}$$

the even (commuting) and odd (anticommuting) subsets of the algebra. It is easy to show that these are subspaces (by linearity of commutators and anticommutators), and **one can also show** that $\mathcal{G}_N$ can be decomposed as

$$\mathcal{G}_N \cong \mathcal{G}_N^+ \oplus \mathcal{G}_N^-.$$

Odd elements satisfy a really important property. If $\psi \in \mathcal{G}_N^+$ it follows that $\{\psi, \psi\} = 2\psi^2 = 0$. That is, all powers greater than one of an odd element vanish.

Definition 2.1. Given $\mathcal{G}$ a Grassmann algebra, as defined above, and $f : \mathbb{C}^k \rightarrow \mathbb{C}$ an analytic function given by its Taylor series centered at zero

$$f(\mathbf{z}) = \sum_{j=0}^{\infty} \sum_{|\alpha|=j} \frac{\partial^\alpha f(0)}{j!} \mathbf{z}^i$$

we define the associated *function of Grassmann variables* as

$$f(\boldsymbol{\psi}) = \sum_{j=0}^{\infty} \sum_{|\alpha|=j} \frac{\partial^\alpha f(0)}{j!} \boldsymbol{\psi}^i$$

where $\boldsymbol{\psi} \in \mathcal{G}_N \times \dots \times \mathcal{G}_N$ is the list of k Grassmann variables.

Remark 16. If $\psi \in \mathcal{G}_N^+$ it follows that any function of Grassmann variables evaluated in ψ has at most first order terms, as all higher orders will always contain a factor of $\psi^2 = 0$. The most important example to us is the exponential of an odd element:

$$e^\psi = 1 + \psi + \frac{1}{2}\psi^2 + \dots = 1 + \psi$$

Remark 17. We could define a Grassmann algebra more "directly" as the ring of polynomials in anticommuting variables over a field, and choose $\mathbb{C}$ as base field to get exactly the structure we need. Indeed, the Grassmann algebra can be viewed as the free algebra generated by the variables corresponding to the modes of the physical system, quotiented by the ideal generated by anticommutators, i.e.

$$\mathcal{G}_N := \frac{\mathbb{C}\langle \psi_1, \dots, \psi_N, \bar{\psi}_1, \dots, \bar{\psi}_N \rangle}{\text{span}\{\psi_i \psi_j + \psi_j \psi_i, \psi_i \bar{\psi}_j + \bar{\psi}_j \psi_i, \bar{\psi}_i \bar{\psi}_j + \bar{\psi}_j \bar{\psi}_i : i, j = 1, \dots, N\}}$$

We now introduce integrals of functions of Grassmann variables. Note that the definition is going to be purely algebraic and, in particular, motivated by our need to reproduce specific computational rules. Indeed, we'll see that we can define expectations of functions of Grassmann variables in a way that reproduces exactly Wick's rule for expectations of fermionic field operators.

We ask, first of all, for the integral to be a linear operator on a Grassmann algebra. Then, specifying the image of a basis for the algebra will uniquely determine the operator.

Definition 2.2. Given $\{1, \psi_1, \dots, \psi_N, \bar{\psi}_1, \dots, \bar{\psi}_N\}$ the basis for the Grassmann algebra, as above, the *Grassmann integral* is the linear operator that acts on this basis as

$$\begin{aligned} \int d\psi_i \psi_j &= \int d\bar{\psi}_i \bar{\psi}_j := \delta_{ij}, \\ \int d\psi_i &= \int d\bar{\psi}_i := 0. \end{aligned}$$

Repeated integration is denoted by a single integral symbol and multiple "differentials".

Example 1. In $\mathcal{G}_2$, consider the exponential function. Using the definition, we can evaluate the following integrals to see how this operator behaves.

- $\int d\psi_1 e^{\psi_1 \bar{\psi}_2} = \int d\psi_1 (1 + \psi_1 \bar{\psi}_2) = \bar{\psi}_2;$
- $\int d\bar{\psi}_2 e^{\psi_1 \bar{\psi}_2} = \int d\bar{\psi}_2 (1 - \bar{\psi}_2 \psi_1) = -\psi_1;$
- $\int d\psi_1 d\bar{\psi}_2 e^{\psi_1 \bar{\psi}_2} = -1.$

Remark 18. Repeated integration, i.e. the composition of multiple integral operators, inherits some form of anticommutativity when acting on the basis elements. Indeed in $\mathcal{G}_1$, for example, we have

$$\begin{aligned}\int d\psi d\bar{\psi} \psi \bar{\psi} &= - \int d\psi d\bar{\psi} \bar{\psi} \psi = -1 \\ &= - \int d\bar{\psi} d\psi \psi \bar{\psi}.\end{aligned}$$

And given the definition of Grassmann functions, this property will be quite useful to us.

Having defined integration, we can now introduce a *Gaussian measure* (in a purely algebraic sense). To make the notation lighter, we define a "dot product", without any geometric meaning, as $\psi \cdot \phi := \sum_{i=1}^k \psi_i \phi_i$.

Definition 2.3. $G \in \text{Mat}(N \times N, \mathbb{C})$ is said to be a *Grassmann covariance* if $G \in \text{GL}(N, \mathbb{C})$.

Lemma 2.1. Let G be a Grassmann covariance, $\psi = (\psi_1, \dots, \psi_N)$ and $\bar{\psi} = (\bar{\psi}_1, \dots, \bar{\psi}_N)$. Then

$$\int \left(\prod_{i=1}^N d\bar{\psi}_i d\psi_i \right) e^{-\bar{\psi} \cdot G^{-1} \psi} = \det(G^{-1})$$

Proof. We carry out the calculation explicitly.

$$\begin{aligned}\int \left(\prod_{i=1}^N d\bar{\psi}_i d\psi_i \right) e^{-\bar{\psi} \cdot G^{-1} \psi} &= \int \left(\prod_{i=1}^N d\bar{\psi}_i d\psi_i \right) \frac{(-1)^N}{N!} \left(\sum_{i,j=1}^N \bar{\psi}_i (G^{-1})_{ij} \psi_j \right)^N \\ &= \int \left(\prod_{i=1}^N d\bar{\psi}_i d\psi_i \right) \frac{1}{N!} \left(\sum_{i,j=1}^N \bar{\psi}_i (G^{-1})_{ij} \psi_j \right)^N \\ &= \int \left(\prod_{i=1}^N d\bar{\psi}_i d\psi_i \right) \frac{1}{N!} \sum_{\sigma, \tau \in S_N} \prod_{j=1}^N (G^{-1})_{\sigma(j), \tau(j)} \bar{\psi}_{\sigma(j)} \psi_{\tau(j)} = \star\end{aligned}$$

Where S_N is the set of permutations of N elements. We can split the product into a product of matrix elements times a product of Grassmann variables, then we use that

$$\prod_{j=1}^N \bar{\psi}_{\sigma(j)} \psi_{\tau(j)} = (-1)^{\text{sgn}(\sigma) + \text{sgn}(\tau)} \prod_{j=1}^N \bar{\psi}_j \psi_j$$

to find that

$$\begin{aligned}\star &= \frac{1}{N!} \sum_{\sigma, \tau \in S_N} (-1)^{\text{sgn}(\sigma) + \text{sgn}(\tau)} \prod_{j=1}^N (G^{-1})_{\sigma(j), \tau(j)} \\ &= \frac{1}{N!} \sum_{\sigma \in S_N} (-1)^{\text{sgn}(\sigma)} \prod_{j=1}^N (G^{-1})_{j, \sigma(j)} \\ &= \det(G^{-1})\end{aligned}$$

□

Definition 2.4. Let G be a Gaussian covariance, and $\{\psi_1, \dots, \psi_N, \bar{\psi}_1, \dots, \bar{\psi}_N\}$ the usual Grassmann variables. A *Grassmann Gaussian measure* is the Grassmann integral measure defined by

$$P_G(d\psi) := \frac{1}{\det(G^{-1})} \left(\prod_{j=1}^N d\bar{\psi}_j d\psi_j \right) e^{-\bar{\psi} \cdot G^{-1} \psi}$$

Remark 19. The factor in front of the differentials guarantees a "normalized" measure, i.e. $\int P_G(d\psi) = (1/\det(G^{-1}))\det(G^{-1}) = 1$.

We will prove the following statement later.

Lemma 2.2. *With the same notation used so far,*

$$\int P_G(d\psi) \psi_i \bar{\psi}_j = G_{ij}$$

Lemma 2.3. *Let $D \in \text{GL}(2, \mathbb{C})$, φ and η two vectors of Grassmann variables. Then, under the following change of variables*

$$\begin{pmatrix} \psi \\ \bar{\psi} \end{pmatrix} = D \begin{pmatrix} \varphi \\ \bar{\varphi} \end{pmatrix} + \begin{pmatrix} \eta \\ \bar{\eta} \end{pmatrix}$$

the integral of a function of Grassmann variables f changes as follows

$$\int \left(\prod_i d\bar{\psi}_i \psi_i \right) f(\psi, \bar{\psi}) = \det(G^{-1}) \int \left(\prod_i d\bar{\varphi}_i \varphi_i \right) f(D(\varphi, \bar{\varphi}) + (\eta, \bar{\eta}))$$

Proof. check proof in the notes (D is defined only in the diagonal NxN blocks? How does he calculate "differentials" of the new variables?). $\square$

Lemma 2.4. *Let $G \in \text{GL}(N, \mathbb{C})$, then*

$$\int P_G(d\psi) e^{\bar{\eta} \cdot \psi + \bar{\psi} \cdot \eta} = e^{\bar{\eta} \cdot G \eta}$$

Proof. it was an exercise. $\square$

Before defining expectations, we need one more operator on a Grassmann algebra to mimic the behaviour of these same integrals done in the complex variables case. That is, we need to introduce derivatives of Grassmann functions. If we had a polynomial of degree N in N variables, then deriving this polynomial in all of the variables should return the maximal degree term. Then this becomes a natural choice for the definition of Grassmann derivatives, as we defined functions to be nothing more than Taylor series, and evaluating a function in the elements of the basis gets us exactly in this situation. But recall that integrating a Grassmann function in all variables (i.e., in all the basis elements) returns exactly the same thing, the coefficient of the maximal degree term. Therefore, this natural definition for Grassmann derivatives reads as follows.

Definition 2.5. Let $f(\psi, \bar{\psi})$ be a Grassmann function, the *Grassmann derivatives* of f w.r.t. the basis elements are the linear operators defined as

$$\frac{\partial}{\partial \psi_i} f := \int d\psi_i f(\psi, \bar{\psi})$$

Proof (Lemma 2.2). Recalling the "anticommutativity property" of Grassmann integrals, it is clear that the same property holds for derivatives too. We can make use of Grassmann derivatives to prove this lemma in a way that is closer to the complex variables case, and this motivates the introduction of Grassmann derivatives. Note that, given a vector of Grassmann variables η ,

$$\psi_i \bar{\psi}_j e^{-\bar{\psi} \cdot G^{-1} \psi} = -\frac{\partial}{\partial \eta_i} \frac{\partial}{\partial \eta_j} e^{-\bar{\psi} \cdot G^{-1} \psi + \bar{\eta} \cdot \psi + \bar{\psi} \cdot \eta} \Big|_{\eta = \bar{\eta} = 0}.$$

We can substitute this in the integral to get that

$$\begin{aligned}
\int P_G(d\psi) \psi_i \bar{\psi}_j &= -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} \int P_G(d\psi) e^{\bar{\eta} \cdot \psi + \bar{\psi} \cdot \eta} \Big|_0 \\
&= -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} e^{\bar{\eta} \cdot G \eta} \Big|_0 \\
&= -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} (\bar{\eta}_i G_{ij} \eta_j) \Big|_0 \\
&= G_{ij}
\end{aligned}$$

which is exactly the thesis. $\square$

With this useful lemma proven, we get to the most important theorem of this section: Wick's theorem. It is this result that allows us to actually interpret Grassmann Gaussian integrals as "expectations of fermionic field operators", because it shows that these integrals exactly reproduce the combinatorics of fermionic expectations.

Theorem 2.5 (Wick's rule). *Let $M \in \mathbb{N}$ be the degree of a monomial in Grassmann variables, and let $I = \{a_1, \dots, a_M\}$ and $J = \{b_1, \dots, b_M\}$ be two sets of M indices. The integral of a monomial is*

$$\int P_G(d\psi) \left(\prod_{\alpha \in I, \beta \in J} \psi_\alpha \bar{\psi}_\beta \right) = \sum_{\sigma \in S_N} (-1)^{\text{sgn}(\sigma)} \prod_{j=1}^M G_{a_j, b_{\sigma(j)}}$$

Proof. check proof of Isserlis' theorem, and check the way the product on the LHS is written. $\square$

I REWORKED THE STRUCTURE OF THIS SECOND CHAPTER! It follows that calling a Grassmann Gaussian integral a "fermionic expectation" makes sense. Our task now is to give important results that greatly simplify the calculation of such expectations. In particular, the aim is to represent these calculations using Feynman diagrams, which are a tool to perform the combinatorics of this exact type of problem, and because these integrals are supposed to be a representation of expectations of physical quantities, Feynman diagrams will have a clear physical interpretation that we will discuss in due time.

The results we are about to give concern both bosonic and fermionic systems, and the differences often lie only in sign changes between expressions. In fact, the rules behind Feynman diagrams are going to be the same, what will change however is the precise formula to evaluate such diagrams in a complete calculation.

2.4 Fermionic expectations

2.5 Perturbative series and Feynman diagrams

3 Explicit Calculation at Second Order

3.1 Duhamel's formula

Motivate the study of this system.

The state space for a system of N identical fermions in a box of side length L with periodic boundary conditions is the space of completely antisymmetric L^2 -functions on the torus $\mathbb{T}^{3N} = (\mathbb{R}/2\pi\mathbb{Z})^{3N}$:

$$L_a^2(\mathbb{T}^{3N}) := \{\psi \in L^2(\mathbb{T}^{3N}) : \psi(x_{\sigma(1)}, \dots, x_{\sigma(N)}) = \text{sgn}(\sigma) \psi(x_1, \dots, x_N) \forall \sigma \in S_N\}$$

The Hamiltonian for weakly interacting particles can be written as a sum of the kinetic energy terms plus a sum of two-body interaction terms:

$$H = \hbar^2 \sum_{i=1}^N (-\Delta_{x_i}) + \lambda \sum_{1 \leq i < j \leq N} V(x_i - x_j)$$

where $V : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a smooth function with period L in all variables and such that $V(x_i - x_j) = V(|x_i - x_j|)$, in particular this means that $\hat{V}(-k) = \hat{V}(k)$, and this fact will become useful in evaluating the second order perturbation to the ground state energy of the system. The scaling factor $\lambda \in \mathbb{R}$ is assumed to be small. Working in the grand canonical ensemble requires to lift this operator to the fermionic Fock space, and in second quantization formalism this means that the Hamiltonian will be written as

$$H = H_0 + V = \sum_{k \in \mathbb{Z}^3} \hbar^2 |k|^2 a_k^* a_k + \sum_{k, p, q \in \mathbb{Z}^3} \lambda \hat{V}(k) a_{p-k}^* a_{q+k}^* a_q a_p$$

where $\hat{V}(k)$ are the Fourier coefficients of V and $a_k^\#$ are creation/annihilation operators on the Fock space that create/annihilate states of definite momentum $k \in \mathbb{Z}^3$.

Since we want to compute the second order contribution to the ground state energy, we will calculate the free energy of the system and take its limit as the temperature goes to zero, that is $\beta \rightarrow \infty$. Recall that at thermal equilibrium the expectation value of any observable is given by the Gibbs state functional

$$\langle \cdot \rangle = \frac{\text{Tr}(\cdot e^{-\beta H})}{\text{Tr}(e^{-\beta H})}.$$

Because the free energy is given by the logarithm of the partition function, we can make use of a useful formula that allows to truncate the quantity $\log(\text{Tr}(e^{-\beta H}))$ to second order contributions in the scaling factor λ .

Proposition 3.1. *Given $H = H_0 + V$ as defined above, the following approximation holds up to terms of order $O(\dots)$ (consider putting λ outside of V , i.e. $H = H_0 + \lambda V$, to make this explicit).*

$$\log \text{Tr} e^{-\beta H} = \log \text{Tr} e^{-\beta H_0} - \beta \langle V \rangle_{H_0} \int_0^\beta d\tau \int_0^\tau d\tau' (\langle V(\tau') V(\tau) \rangle_{H_0} - \langle V(\tau') \rangle_{H_0} \langle V(\tau) \rangle_{H_0}) \quad (8)$$

Proof. Start by rewriting the operator $e^{-\beta H} e^{\beta H_0}$ using the fundamental theorem of calculus:

$$\begin{aligned} e^{-\beta H} e^{\beta H_0} &= \mathbb{1} + \int_0^\beta d\tau \left(\frac{d}{d\tau} e^{-\tau H} e^{\tau H_0} \right)(\tau) \\ &= \mathbb{1} + \int_0^\beta d\tau e^{-\tau H} (-H + H_0) e^{\tau H_0} \\ &= \mathbb{1} + \int_0^\beta d\tau e^{-\tau H} e^{\tau H_0} e^{-\tau H_0} (-V) e^{\tau H_0} \\ &= \mathbb{1} - \int_0^\beta d\tau e^{-\tau H} e^{\tau H_0} V(\tau) = \star \end{aligned}$$

The integral converges because of Stone's theorem on strongly continuous unitary groups. Now we can see that in the integral we have the same term that appears in the l.h.s. but at a generic temperature τ . This means that we can repeat the same steps to push this operator to higher and

higher "orders". Iterating this procedure just once gives the following:

$$\begin{aligned}\star &= \mathbb{1} - \int_0^\beta d\tau \left(\mathbb{1} + \int_0^\tau d\tau' e^{-\tau'H} e^{\tau'H_0} V(\tau') \right) V(\tau) \\ &= \mathbb{1} - \int_0^\beta d\tau V(\tau) + \int_0^\beta d\tau \int_0^\tau d\tau' V(\tau') V(\tau) + R(V, \beta).\end{aligned}$$

where $R(V, \beta)$ is the remainder of this expression **and is of order $O(\dots)$** . Dropping the remainder term and truncating the expression, we can invert this relation to isolate $e^{-\beta H}$ and take the trace. Since the trace is a series of matrix elements **we can move it inside the integral** to get:

$$\begin{aligned}\text{Tr } e^{-\beta H} &= \text{Tr } e^{-\beta H_0} - \int_0^\beta d\tau \text{Tr}(V(\tau) e^{-\beta H_0}) + \int_0^\beta d\tau \int_0^\tau d\tau' \text{Tr}(V(\tau') V(\tau) e^{-\beta H_0}) \\ &= (\text{Tr } e^{-\beta H_0}) \left(1 - \int_0^\beta d\tau \langle V(\tau) \rangle_{H_0} + \int_0^\beta d\tau \int_0^\tau d\tau' \langle (V(\tau') V(\tau)) \rangle_{H_0} \right).\end{aligned}$$

We can now take the logarithm on both sides. Recall the Taylor expansion $\log(1+x) \sim x - x^2/2$ as $x \rightarrow 0$ and apply it to the r.h.s., but note that the first integral gives two contributions (one at first order and one at second order) while the second integral only contributes to the second order. In the end, we are left with:

$$\begin{aligned}\log \text{Tr } e^{-\beta H} &= \log \text{Tr } e^{-\beta H_0} - \int_0^\beta d\tau \langle V(\tau) \rangle_{H_0} \\ &\quad + \int_0^\beta d\tau \int_0^\tau d\tau' \langle (V(\tau') V(\tau)) \rangle_{H_0} - \frac{1}{2} \left(\int_0^\beta d\tau \langle V(\tau) \rangle_{H_0} \right)^2 \\ &= \log \text{Tr } e^{-\beta H_0} - \beta \langle V \rangle_{H_0} \\ &\quad + \int_0^\beta d\tau \int_0^\tau d\tau' \left(\langle V(\tau') V(\tau) \rangle_{H_0} - \langle V(\tau') \rangle_{H_0} \langle V(\tau) \rangle_{H_0} \right)\end{aligned}\tag{9}$$

where in the last step we performed the first integral and rewrote the square as a double integral, then taking into account the symmetry under $\tau \longleftrightarrow \tau'$ we could remove the factor 1/2 by integrating only over the lower triangle $\{(\tau, \tau') \in [0, \beta]^2 : \tau' \leq \tau\}$. $\square$

3.2 Evaluating expectation values

To evaluate the expectation values, which are expectations of polynomials in field operators, we can turn to Grassmann variables. Indeed, **by introducing a cutoff Λ** we see that the possible momentum states form a finite set of M independent modes:

$$\mathcal{D}_\Lambda := \{\dots\}.$$

Let then $\psi = (\psi_1, \dots, \psi_M)$ and $\bar{\psi} = (\bar{\psi}_1, \dots, \bar{\psi}_M)$ be the Grassmann variables associated to the families of operators a and a^* respectively. Having introduced the cutoff, the Hamiltonian can now be written as a simple finite linear combination of creation/annihilation operators:

$$H_0 = \sum_{k \in \mathcal{D}_\Lambda} \epsilon_k a_k^* a_k$$

We begin the calculation by building the gaussian measure for the expectations $\langle \cdot \rangle_{H_0}$. Note that $\langle V \rangle_{H_0}$ contains field operators at the same temperature, while the expectations inside the double integral contain field operators at different temperatures, which means the two-point functions that appear in the contractions of the respective diagrams are going to be different.

Lemma 3.2. *The two-point function for this system has the following form:*

$$G_{k,k'}(\tau, \tau') = \langle a_k^*(\tau) a_{k'}(\tau') \rangle_{H_0} = \delta_{k,k'} n_F(\epsilon_k) e^{-(\tau - \tau') \epsilon_k} \quad (10)$$

where $n_F(\epsilon) = (e^{\beta\epsilon} + 1)^{-1}$ is the Fermi-Dirac distribution.

Proof. We are going to divide the proof in two parts: first we will find an expression for the operators $a^\#(\tau)$, as this will allow us to write the two-point function in terms of $\langle a_k^* a_k' \rangle$, then we will compute this last expectation to find the Fermi-Dirac distribution. Recall that in the Heisenberg picture with imaginary time one has that for an operator A its evolution at temperature τ is given by $A(\tau) = e^{-\tau H_0} A e^{\tau H_0}$. **check domains**. Once again we will make use of the fundamental theorem of calculus to write

$$\begin{aligned} e^{-\tau H_0} a_k e^{\tau H_0} &= a_k + \int_0^\tau d\lambda \frac{d}{d\lambda} (e^{-\lambda H_0} a_k e^{\lambda H_0}) \\ &= a_k + \int_0^\tau d\lambda e^{-\lambda H_0} (-H_0 a_k + a_k H_0) e^{\lambda H_0} \\ &= a_k + \int_0^\tau d\lambda e^{-\lambda H_0} [a_k, H_0] e^{\lambda H_0} \\ &= a_k + \int_0^\tau d\lambda e^{-\lambda H_0} \sum_{k' \in \mathcal{D}_\Lambda} \epsilon_{k'} [a_k, a_{k'}^* a_{k'}] e^{\lambda H_0} = \star. \end{aligned}$$

Recall that, for operators A, B and C defined on a common domain, it holds that

$$[A, BC] = \{A, B\}C - B\{A, C\}$$

as can be shown by explicitly writing the expressions for the anticommutators. Using this identity we rewrite the commutator that appears in the last expression in terms of the canonical anticommutator as follows:

$$\begin{aligned} \star &= a_k + \int_0^\tau d\lambda e^{-\lambda H_0} \sum_{k' \in \mathcal{D}_\Lambda} \epsilon_{k'} (\{a_k, a_{k'}^*\} a_{k'} - a_{k'}^* \{a_k, a_{k'}\}) e^{\lambda H_0} \\ &= a_k + \int_0^\tau d\lambda e^{-\lambda H_0} \epsilon_k a_k e^{\lambda H_0} \\ &= a_k + \epsilon_k \int_0^\tau d\lambda a_k(\lambda) \end{aligned}$$

where we used that $\{a_k, a_{k'}^*\} = \delta_{k,k'}$ and that $\{a_k, a_{k'}\} = 0$. We now have an expression for $a_k(\tau)$ in terms of its integral, and if we take the derivative with respect to τ we get the following Cauchy problem:

$$\begin{cases} \frac{d}{d\tau} a_k(\tau) = \epsilon_k a_k(\tau) \\ a_k(0) = a_k \end{cases}.$$

By the **theorem of existence and uniqueness of the solution to an ODE on a closed interval**, it suffices to find one explicitly. This is a well known differential equation and is easy to check that the following is indeed a solution

$$a_k(\tau) = e^{\tau \epsilon_k} a_k(0) = e^{\tau \epsilon_k} a_k.$$

The procedure is completely analogous in the case of $a_k^*(\tau)$ and the result differs only by a sign in the exponent:

$$a_k^*(\tau) = e^{-\tau \epsilon_k} a_k^*(0) = e^{-\tau \epsilon_k} a_k^*.$$

By linearity, we can take the scalar exponential factors out of the expectation value to write the two-point function in terms of the quantity $\langle a_k^* a_{k'} \rangle$:

$$\langle a_k^*(\tau) a_{k'}(\tau') \rangle_{H_0} = e^{-\tau \epsilon_k} e^{\tau' \epsilon_{k'}} \langle a_k^* a_{k'} \rangle_{H_0}.$$

It remains to determine the expectation $\langle a_k^* a_{k'} \rangle$ and find the Fermi-Dirac distribution. Recall that the spectrum of the number density operators $a_k^* a_k$ is $\mathbb{N}_0$, and in the fermionic case the occupation number basis decomposes the Fock space like we discussed in Proposition 1.12. Using the cutoff on high momentum, the trace in the Gibbs state functional becomes a sum over a finite amount of states, therefore let $\{|\mathbf{n}\rangle = |n_1\rangle \otimes \cdots \otimes |n_M\rangle : n_i \in \{0, 1\} \forall i = 1, \dots, M\}$ be the complete orthonormal set of eigenvectors of M independent number density operators (in our case, $M = |\mathcal{D}_\Lambda|$). Calling $Z = \text{Tr} e^{-\beta H_0}$ and using the momenta $k \in \mathcal{D}_\Lambda$ as indices for the basis states labels n_k , we write

$$\begin{aligned} \langle a_k^* a_{k'} \rangle_{H_0} &= \frac{\text{Tr}(a_k^* a_{k'} e^{-\beta H_0})}{\text{Tr} e^{-\beta H_0}} = \frac{1}{Z} \sum_{\mathbf{n} \in \{0, 1\}^M} \langle \mathbf{n} | e^{-\beta H_0} a_k^* a_{k'} | \mathbf{n} \rangle \\ &= \delta_{k, k'} \frac{1}{Z} \sum_{\mathbf{n} \in \{0, 1\}^M} \langle \mathbf{n} | a_k^* a_k | \mathbf{n} \rangle \prod_{p \in \mathcal{D}_\Lambda} e^{-\beta \epsilon_p n_p} \\ &= \delta_{k, k'} \frac{1}{Z} \sum_{\mathbf{n} \in \{0, 1\}^M} n_k e^{-\beta \epsilon_k n_k} \prod_{p \neq k} e^{-\beta \epsilon_p n_p} \\ &= \delta_{k, k'} \frac{1}{Z} \left(\sum_{n_k=0}^1 n_k e^{-\beta \epsilon_k n_k} \right) \prod_{p \neq k} \sum_{n_p=0}^1 e^{-\beta \epsilon_p n_p} \\ &= \delta_{k, k'} \frac{1}{Z} e^{-\beta \epsilon_k} \prod_{p \neq k} (1 + e^{-\beta \epsilon_p}) \end{aligned}$$

where we used the fact that $e^{-\beta H_0}$ is **bounded** and diagonal in the occupation number basis $\{|\mathbf{n}\rangle\}$, and that $a_k^* a_k |\mathbf{n}\rangle = n_k |\mathbf{n}\rangle$. Using many of the same steps for the denominator, we find that $Z = \prod_{p \in \mathcal{D}_\Lambda} (1 + e^{-\beta \epsilon_p})$, thus the expression simplifies to

$$\begin{aligned} \langle a_k^* a_{k'} \rangle_{H_0} &= \delta_{k, k'} \frac{e^{-\beta \epsilon_k} \prod_{p \neq k} (1 + e^{-\beta \epsilon_p})}{\prod_p (1 + e^{-\beta \epsilon_p})} \\ &= \delta_{k, k'} \frac{e^{-\beta \epsilon_k}}{1 + e^{-\beta \epsilon_k}} \\ &= \delta_{k, k'} \frac{1}{e^{\beta \epsilon_k} + 1}. \end{aligned}$$

This is exactly the Fermi-Dirac distribution, and the thesis follows immediately. $\square$

With this quantity we can go over to the computation via Feynman diagrams. Recall the definition for the expectation of a function of Grassmann variables

$$\mathbb{E}_G(\cdot) = \int P_G(d\psi)(\cdot), \quad \text{where } P_G(d\psi) = \frac{1}{\det G} \left(\prod_{j=1}^N d\psi_j d\bar{\psi}_j \right) e^{-\bar{\psi} G^{-1} \psi}$$

and the graphical representation of the interaction term via a vertex:

$$V = \lambda \sum_{k, p, q} \hat{V}(k) \bar{\psi}_{p-k} \bar{\psi}_{q+k} \psi_q \psi_p = \begin{array}{c} \text{Diagram: A central vertex with four external lines. Top-left line labeled } p-k \text{ (incoming), top-right line labeled } q+k \text{ (incoming), bottom-left line labeled } q \text{ (incoming), bottom-right line labeled } p \text{ (incoming).} \end{array}.$$

With these we can evaluate the expectations that show up in Duhamel's formula separately. Let's start with $\langle V \rangle_{H_0}$. This is the expectation of a sum with terms $\bar{\psi}\psi\psi\psi$, exactly the interaction term, and so the result will be a sum of all possible (connected) diagrams with one vertex. Omitting the temperatures when writing two-point functions and calling $n_k = n_F(\epsilon_k)$ the calculation is as follows:

$$\begin{aligned}
\langle V \rangle_{H_0} &= \mathbb{E}_G \left(\lambda \sum_{k,p,q} \hat{V}(k) \bar{\psi}_{p-k} \bar{\psi}_{q+k} \psi_q \psi_p \right) = \sum_{\Gamma \in \mathcal{G}_{C,1}} \text{val}(\Gamma) \\
&= \text{draw feynman diagrams} \\
&= \sum_{k,p,q} \hat{V}(k) G_{p-k,p} G_{q+k,q} - \sum_{k,p,q} \hat{V}(k) G_{p-k,q} G_{q+k,p} \\
&= \sum_{k,p,q} (\hat{V}(k) \delta_{k,0} n_p n_q - \hat{V}(k) \delta_{k,p-q} n_p n_q) \\
&= \sum_{p,q} (\hat{V}(0) - \hat{V}(p-q)) n_p n_q
\end{aligned}$$

Let's consider the term $\langle V(\tau') V(\tau) \rangle - \langle V(\tau') \rangle \langle V(\tau) \rangle$ in the double integral now. This time we have terms with two vertices, and recalling the linked cluster theorem the expression factorizes as follows:

$$\begin{aligned}
\langle V(\tau') V(\tau) \rangle - \langle V(\tau') \rangle \langle V(\tau) \rangle &= \mathbb{E}_G(V(\tau') V(\tau)) - \mathbb{E}_G(V(\tau')) \mathbb{E}_G(V(\tau)) \\
&= \mathbb{E}_G^T(V(\tau'); V(\tau)) \\
&= \sum_{\Gamma \in \mathcal{G}_{C,2}} \text{val}(\Gamma) = \star
\end{aligned}$$

Assuming that V is such that $\hat{V}(-k) = \hat{V}(k)$, the connected diagrams with two vertices are exactly divided in two groups, up to renaming the indices in the resulting sums. Indeed, on one hand we have diagrams that have no loops, i.e. each vertex only has contractions with the other vertex and never with itself, and on the other we have diagrams with two loops.

draw the complete set of diagrams?

Consider the first group. Fixing an incoming and an outgoing leg on one vertex, there are 4 ways to contract them with legs of the other vertex, and in each case there is only one way to complete the diagram without introducing loops, therefore this group contains 4 diagrams. For the second group, there are 4 ways to form a loop on a vertex, and given a (connected) diagram with two vertices and two loops there exists only one way to complete it. Therefore the second group contains 16 diagrams (all possible ways to put one loop on both vertices). Accounting for these multiplicities, we have:

$$\begin{aligned}
\star &= 4 (\text{noloops}) + 16 (\text{withloops}) \\
&= \sum_{k,p,q,k',p',q'} \hat{V}(k') \hat{V}(k) (4 G_{p-k,p'} G_{q'+k',q} G_{q+k,q'} G_{p'-k',p} + 16 G_{p-k,q} G_{q+k,q'} G_{p'-k',p} G_{q'+k',p'}) \\
&= \sum_{p,q,p'} \left(4 (\hat{V}(p-p'))^2 e^{(\tau'-\tau)(\epsilon_p+\epsilon_q+\epsilon_{p'}+\epsilon_{p-p'+q})} n_p n_q n_{p'} n_{p-p'+q} \right. \\
&\quad \left. + 16 \hat{V}(0) \hat{V}(p-q) e^{(\tau'-\tau)(2\epsilon_p+\epsilon_q+\epsilon_{p'})} n_p^2 n_q n_{p'} \right)
\end{aligned}$$

Renaming some of the indices at both orders and integrating this last expression, as per equation (9), leads to the final expression of the free energy at second order

$$\begin{aligned}
\log \text{Tr } e^{-\beta H} &= \log \text{Tr } e^{-\beta H_0} \\
&- \beta \sum_{p,k \in \mathcal{D}_\Lambda} \left(\hat{V}(0) - \hat{V}(k) \right) n_p n_{p-k} \\
&- \sum_{p,q,k \in \mathcal{D}_\Lambda} \left(\frac{4(\hat{V}(k))^2 n_p n_q n_{q+k} n_{p-k}}{\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k}} \left(\beta + \frac{1 - e^{-\beta(\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k})}}{\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k}} \right) \right. \\
&\quad \left. + \frac{16\hat{V}(0)\hat{V}(p-q)n_p^2 n_q n_{p-k}}{2\epsilon_p + \epsilon_q + \epsilon_{p-k}} \left(\beta + \frac{1 - e^{-\beta(2\epsilon_p + \epsilon_q + \epsilon_{p-k})}}{2\epsilon_p + \epsilon_q + \epsilon_{p-k}} \right) \right).
\end{aligned}$$

Now that the computation is complete, we need to prove that removing the UV cutoff, i.e. taking the limit $\Lambda \rightarrow \infty$, leads to a well-defined expression. Please note that this is not at all a trivial statement and what we are doing is deciding on the order in which to take different limits, as removing the cutoff starting from this expression means that the perturbative series is *not* done in the case where the state space is infinite-dimensional. Physically, this does not pose any concern and we expect the two scenarios to be equivalent, but from the point of view of mathematical rigour it is worth pointing out this kind of subtleties.

Theorem 3.3. *With the regularity condition that, given $m \geq 1$, $\partial^\alpha V \in L^1(\mathbb{T}^3)$ for all multi-indices α such that $|\alpha| = m$, and the assumption that ϵ_p is a positive and monotonically increasing **polynomial** in $|p|$, the above expression converges in the limit $\Lambda \rightarrow \infty$.*

Proof. The zero-th order is trivial, as it doesn't depend on Λ and is, in fact, finite: $e^{-\beta H_0}$ is trace-class and strictly positive definite, which implies $|\log \text{Tr } e^{-\beta H_0}| < \infty$. Moving onto the first and second orders, by the non-stationary phase argument we can conclude that $|\hat{V}(k)| \leq C/|k|^m$ for some $C > 0$, which means that at both orders the factors $\hat{V}(k)$ will be decaying as an inverse power as $|k| \rightarrow \infty$, which is the relevant limit as we take the cutoff Λ to infinity. As for the energy terms, we are assuming that $\epsilon_k \geq 0$ for all k , and that it is a monotonically increasing polynomial in $|k|$. The last factor to discuss is the Fermi-Dirac distribution, but this has an explicit form: an exponential decay in ϵ_k , which leads to an exponential decay in some power (greater or equal to one) of $|k|$.

With these estimates in place we can show that, in the limit $\Lambda \rightarrow \infty$, the expression above converges. Looking at the first order, we can write

$$\begin{aligned}
\sum_{p,k \in \mathcal{D}_\Lambda} \left| \left(\hat{V}(0) - \hat{V}(k) \right) n_p n_{p-k} \right| &\leq |\hat{V}(0)| \sum_{p,k \in \mathcal{D}_\Lambda} n_p n_{p-k} + \sum_{p,k \in \mathcal{D}_\Lambda} |\hat{V}(k)| n_p n_{p-k} \\
&= |\hat{V}(0)| \sum_{p \in \mathcal{D}_\Lambda} n_p \sum_{k \in \mathcal{D}_\Lambda} n_{p-k} + \sum_{p \in \mathcal{D}_\Lambda} n_p \sum_{k \in \mathcal{D}_\Lambda} |\hat{V}(k)| n_{p-k}.
\end{aligned}$$

By the positivity of the general terms for all $k \in \mathbb{Z}^3$ we can bound the sums by the respective sum over $\mathbb{Z}^3$, and because the general terms of each of the above sums only depends on the modulus of the index we can employ the following strategy. Define $A_n = \{k \in \mathbb{Z}^3 : n \leq |k| < n+1\}$ and write

$$\sum_{k \in \mathbb{Z}^3} a_{|k|} = \sum_{n=0}^{\infty} \sum_{k \in A_n} a_{|k|}.$$

Because $c_1 |A_n| a_{n+1} \leq \sum_{k \in A_n} a_{|k|} \leq c_2 |A_{n+1}| a_n$ for all n if a_n is decreasing, and because $|A_n| \asymp n^2$, we can conclude that the behaviour of $\sum_{k \in \mathbb{Z}^3} a_{|k|}$ and $\sum_{n=0}^{\infty} n^2 a_n$ is the same.

Now we can show that each term converges. Indeed $\sum_{k \in \mathbb{Z}^3} n_k$ converges if, and only if, the series $\sum_{\ell \in \mathbb{N}} \ell^2 e^{-\beta \epsilon_\ell}$ converges, by the asymptotic comparison test. The latter does converge absolutely as shown by the root test:

$$\left| \ell^2 e^{-\beta \epsilon_\ell} \right|^{1/\ell} = e^{\frac{2 \log(\ell)}{\ell} - \beta \epsilon_\ell} \longrightarrow K \in [0, 1).$$

Similarly, $\sum_{k \in \mathbb{Z}^3} |\hat{V}(k)| n_k$ converges, as

$$\sum_{k \in \mathbb{Z}^3} |\hat{V}(k)| n_k \leq \sum_{k \in \mathbb{Z}^3} \frac{C n_k}{|k|^m} \leq \sum_{k \in \mathbb{Z}^3} C n_k.$$

This shows that indeed the first order is, in the limit in $\Lambda \rightarrow \infty$, an absolutely convergent series.

Very similar arguments can be used to show that the second order has an absolutely convergent series as a limit. Indeed, recalling the assumptions made about each factor present in the expression, we can bound the first term in the sum by

$$\sum_{p,q,k \in \mathcal{D}_\Lambda} (\dots) \leq \sum_p \frac{4n_p}{\epsilon_p} \sum_q n_q \sum_k (\hat{V}(k))^2 n_{q+k} n_{p-k} \left(\beta + \frac{1}{\epsilon_p} \right),$$

which, turning once more sums over $\mathbb{Z}^3$ into normal series with an additional square in the general term, is an absolutely convergent series in the limit $\Lambda \rightarrow \infty$, as the only true difference from the previous case is the additional factor $1/\epsilon_p$, which is an inverse power and does not change the result of any of the comparison tests done before. The exact same argument can be applied to the second term, and this concludes the proof. $\square$

And with the convergence of the free energy under control, the next step is to look at the limit $\beta \rightarrow \infty$ to retrieve the ground state energy.

3.3 Mean field scaling limit

A More about operators on a Hilbert space

We group in this appendix a series of definitions and theorems that are not necessarily part of the basic foundations of quantum mechanics, but are still crucial in understanding how exactly some computations are carried out and why they hold true. Some of the results presented here go beyond the scope of our work as far as their proof is concerned, and only serve the purpose of painting a clearer picture of what some tools actually are without just referencing an entire course on functional analysis.

Recall that the dual of a complex Hilbert space $\mathcal{H}$ is the Hilbert space $\mathcal{H}^* = \{\varphi : \mathcal{H} \rightarrow \mathbb{C} : \varphi \text{ is a continuous linear functional}\}$, where continuity is with respect to the norm topology on $\mathcal{H}$.

Theorem A.1 (Riesz representation theorem). *Let $\mathcal{H}$ be a complex Hilbert space and $F : \mathcal{H} \rightarrow \mathbb{C}$ a linear functional. Then $F \in \mathcal{H}^*$ if, and only if, there exists a unique $x_F \in \mathcal{H}$ such that $F(x) = \langle x_F, x \rangle$ for all $x \in \mathcal{H}$. Moreover, it holds that $\|x_F\|_{\mathcal{H}} = \|F\|_{\mathcal{H}^*}$.*

Proof. If $\ker(F) = \mathcal{H}$ then $x_F = 0$. Otherwise, let $y \in \ker(F)^\perp$ and define $x_F = \frac{F(y)}{\|y\|^2} y$. $\square$

TTOODDOO:

- introduce UNITARIES and ISOMETRIES (and talk more about self-adjoint operators?)
- ORTHOGONAL COMPLEMENT theorem + RIESZ representation theorem
- SPECTRAL theorem (which version?)
- DIRECT SUMS and TENSOR PRODUCTS of hilbert spaces (and their uniqueness)
 $\longrightarrow$ move N-fold tensor product state space definition/explanation here, and move construction of (anti-)symmetric tensor products to the discussion on canonical/grand canonical pictures, it makes more sense.

- INTEGRALS (and derivatives) of operator-valued functions
- comments on and PROPERTIES of $e^{-\beta H_0}$ (with H_0 bounded and maybe even unbounded)
- a couple of comments on VARIATIONAL PROBLEMS? (from blanchard maybe)

B On particle-hole excitations

How do we apply all this theory to, for example, BCS superconductors - since Bogolubov transformations do not necessarily preserve the relations $[a_{\pm}, a_{\pm}]_{\mp} = 0$ and $[a_{\pm}^*, a_{\pm}^*]_{\mp} = 0$ and we did rely on those too at times.

C References

[1]

References

[1] What. “wHAT”. In: *wHaT* (which year again?).